IDC409 : Introduction to Data Science and Artificial Intelligence

Air Quality Prediction

Group 14

Group Members

Omansh Malhotra	MS23122
Gyandeep Verma	MS23160
Md Anas Rahman	MS23246

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Chapter 1

Project overview: what problem are we solving?

1.1 The practical question

Air pollution is measured through concentrations of several pollutants. A city may have high particulate matter, nitrogen oxides, ozone or other pollutants on a given day. Communicating all concentrations individually is difficult, so an index called the Air Quality Index compresses the readings into a single number and a category such as Good, Moderate or Poor.

This project uses daily, city-level observations from India to investigate three connected questions:

1. **Describe the data:** how do pollution levels vary by city, year, month, season, pollutant and special event?
2. **Estimate and classify AQI:** can machine-learning models infer the numeric AQI or its category from pollutant and calendar information?
3. **Forecast the next day:** can information available today improve on the simple assumption that tomorrow's AQI will equal today's?

A fourth theme is data engineering: the project does not jump directly from a CSV to a model. It creates a SQLite database, applies auditable cleaning steps, constructs a modelling table and stores analysis/model results.

1.2 The end-to-end workflow

Original city-day CSV

⇓

SQLite staging, dimension and fact tables

⇓

Cleaning + audit log + integrity checks

⇓

Gap-aware feature engineering

⇓

EDA + event analysis + feature reduction

⇓

Chronological training / validation / test

⇓

Regression + classification + next-day forecasting

⇓

Metrics, diagnostic figures, interpretation and limitations

Each stage has a purpose. A database gives the observations structure and traceable transformations. Cleaning prevents invalid readings from masquerading as real measurements. Feature engineering converts dates and histories into information a model can use. EDA builds an understanding of the data before fitting models. Chronological evaluation tests performance on later observations rather than a random mixture of past and future.

1.3 What is delivered in the repository?

The archive contains seven executed Jupyter notebooks, reusable Python modules in `src/`, SQL schema and pipeline scripts, the original and processed CSV files, a populated SQLite database, saved scikit-learn pipelines, result CSV/JSON files, 54 figures and a 13-test unit-test suite. The notebooks are ordered from database creation through cleaning, EDA, feature reduction, regression, classification and next-day/city analysis.

The project is not one model. It is a small reproducible data-science system: ingestion, database design, data-quality decisions, statistical exploration, feature design, model training, evaluation and communication.

Chapter 2

Air Quality Index: the domain concept

2.1 Pollutant concentration versus AQI

A pollutant concentration is a measured physical quantity, usually expressed in a concentration unit. Examples include particulate matter in micrograms per cubic metre and carbon monoxide in milligrams per cubic metre. AQI is different: it is a dimensionless index derived from pollutant-specific sub-indices and then mapped to a health-oriented category.

The dataset contains twelve pollutant columns: PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene and Xylene. It also contains the reported numeric AQI and a categorical AQI bucket. The model uses eleven pollutants; Xylene remains in the database but is omitted from modelling because approximately 61% of its values are missing.

2.2 The six CPCB categories

The project uses the six category bands associated with the Indian National AQI convention.

Category	AQI range	Interpretation in broad terms
Good	0–50	Low pollution; minimal health impact expected.
Satisfactory	51–100	Minor breathing discomfort may occur in sensitive people.
Moderate	101–200	Discomfort can affect sensitive groups, including people with lung/heart conditions.
Poor	201–300	Breathing discomfort may affect many people on prolonged exposure.
Very Poor	301–400	Respiratory effects may occur on prolonged exposure.
Severe	> 400	Health effects may occur even in otherwise healthy people; greater risk for vulnerable groups.

The exact wording of health guidance belongs to the official CPCB documentation; this table is a compact educational summary. AQI boundaries matter in this project because classification converts a continuous number into one of these six labels. A prediction close to a boundary can have a small numeric error but a different category.

2.3 How the AQI formula works

The CPCB approach first converts each available pollutant concentration into a pollutant-specific sub-index. Within the relevant concentration interval, the conversion is linear:

$$I_p = \frac{I_{HI} - I_{LO}}{B_{HI} - B_{LO}} (C_p - B_{LO}) + I_{LO}, \tag{2.1}$$

where C_p is the observed concentration of pollutant p, B_{LO} and B_{HI} are the lower and upper concentration breakpoints, and I_{LO} and I_{HI} are the corresponding index breakpoints.

The reported overall AQI is the maximum available pollutant sub-index, subject to a data-completeness condition:

$$\text{AQI} = \max_p I_p. \tag{2.2}$$

The implementation requires at least three valid pollutant sub-indices, including at least one particulate-matter sub-index (PM2.5 or PM10). The code uses seven pollutant sub-indices (PM2.5, PM10, NO2, O3, CO, SO2 and NH3). The highest segment is linearly extrapolated so the reconstructed index can exceed 500, matching the fact that some source values do.

Why take the maximum? AQI is intended to represent the pollutant with the most severe sub-index on that day. It is not an average of all pollutants. If PM2.5 gives a sub-index of 240, PM10 gives 170 and the remaining valid pollutants give lower values, the overall AQI is 240.

2.4 Why this formula matters for the machine-learning task

If the input contains today's pollutant concentrations and the target is today's AQI, then the target is already a mathematical function of much of the input. A model can therefore learn an approximation to the AQI calculation. High same-day accuracy is expected and does not, by itself, demonstrate that the model can forecast air quality in advance.

This leads to a critical distinction throughout the report:

- **Same-day estimation** asks: given today's measured pollutants, how accurately can we reproduce today's reported AQI?
- **Next-day forecasting** asks: before tomorrow's pollutant measurements are available, how accurately can we predict tomorrow's AQI using today's readings and known calendar information?

Chapter 3

Dataset and initial data understanding

3.1 Dataset profile

The main file is `data/raw/city_day.csv`, derived from the Air Quality Data in India collection. It contains 29,531 rows and 16 columns: City, Date, twelve pollutant measurements, AQI and `AQI_Bucket`. The observation period is 1 January 2015 to 1 July 2020 and the dataset covers 26 cities. There are no meteorological variables such as wind speed, rainfall, temperature or boundary-layer height.

Property	Value
Original observations	29,531
Cities	26
Columns	16
Pollutant variables	12
Date range	2015-01-01 to 2020-07-01
Completely empty rows removed	1,374
Rows remaining after SQL cleaning	28,157
Rows eligible for same-day modelling	23,500

A row represents a city on a date. It is not a station-level reading and it is not an hourly record. This distinction affects the interpretation of both the AQI and the model results: city-day records average/aggregate away local variation and cannot capture within-day pollution peaks.

3.2 Missingness is a property of the measurement process

Missing data are not spread evenly across variables. Xylene has the largest missing fraction (about 61.3%), followed by PM10 (37.7%), NH3 (35.0%), Toluene (27.2%), Benzene (19.0%), AQI (15.9%) and PM2.5 (15.6%). The remaining pollutant columns have lower but still meaningful missingness. Missingness also changes over time as monitoring coverage improves.

3.3 Uneven city coverage

The dataset is an unbalanced panel: each city does not contribute the same number of days. Six cities have close to five and a half years of coverage, but Chandigarh has only 304 days beginning in September 2019. Mumbai's AQI is missing for about 61% of its rows. These facts matter because a model can appear accurate by learning the behaviour of cities and periods that dominate the sample. A city-specific comparison should therefore use a shared date window, and a time split should be examined for city coverage as well as overall sample size.

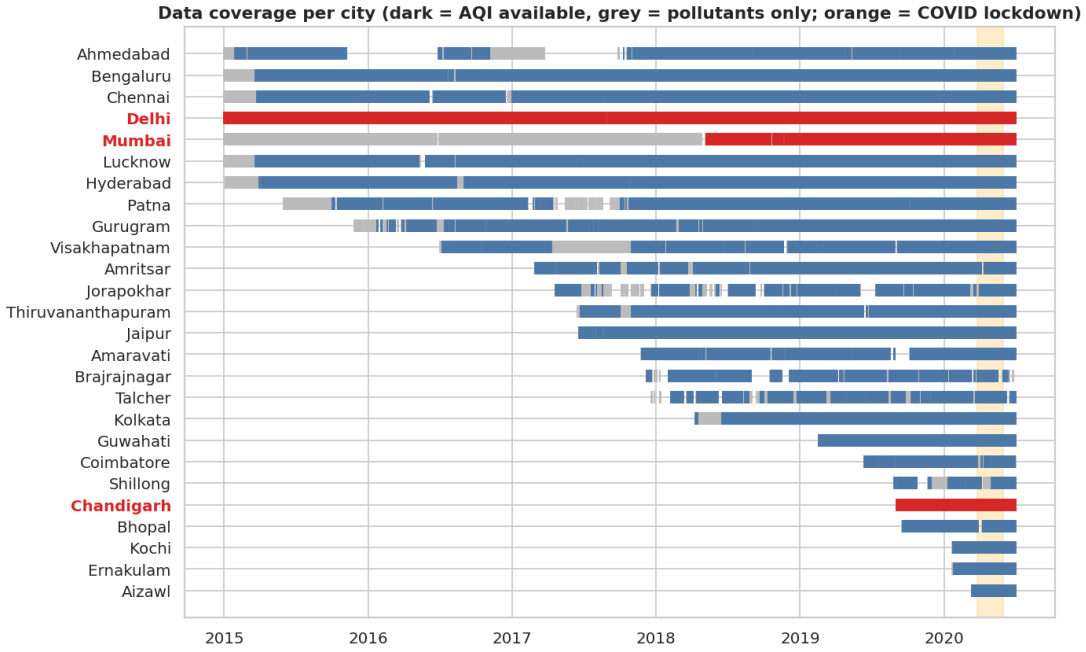


Figure 3.1: Coverage is not uniform across cities. Several cities have almost the full 2015–2020 period, while Chandigarh has only 304 observations from September 2019 to July 2020.

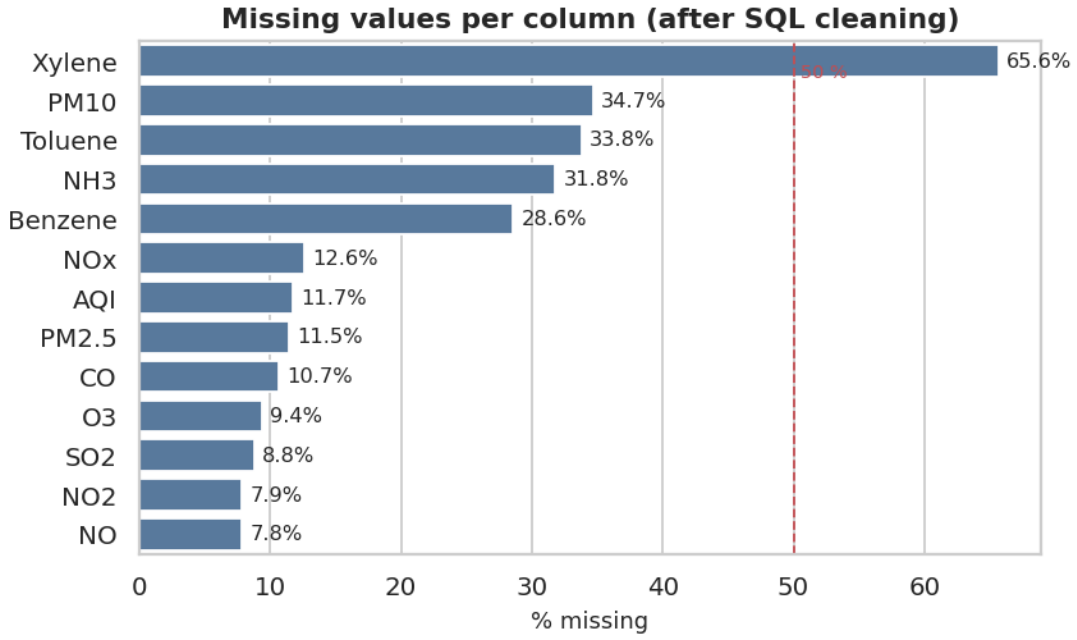


Figure 3.2: Percentage of missing observations by variable. Xylene is especially incomplete and was therefore excluded from model inputs.

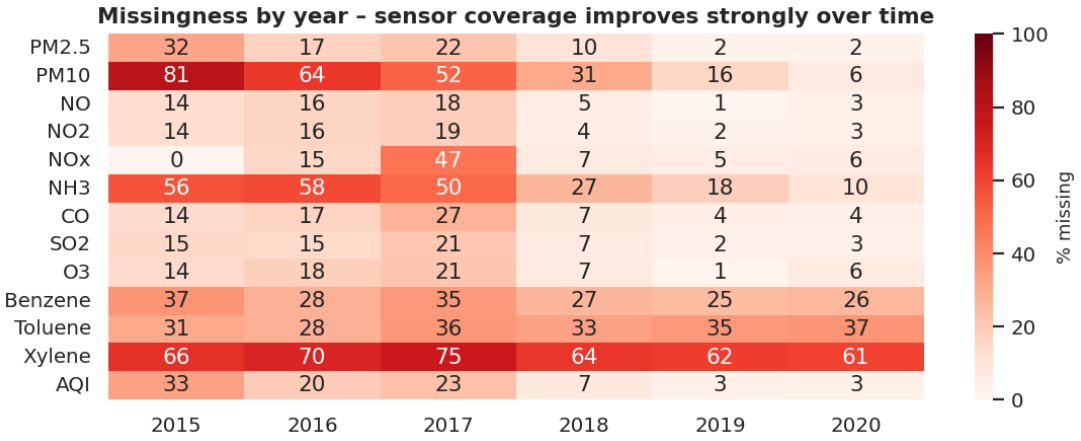


Figure 3.3: Missingness varies across years, so a single overall percentage hides changes in monitoring coverage.



Figure 3.4: Distribution of consecutive missing gaps for PM2.5. Short gaps are common enough to motivate a limited, causal forward-fill rule.

3.4 The unusual Ahmedabad data

Ahmedabad is flagged as a suspect city. Its mean AQI is approximately 452, NO and CO are numerically identical on about 85.2% of days, and it accounts for roughly 76% of observations whose AQI is above 500. These patterns are unusual enough to indicate a possible data-quality or sensor-reporting issue. The project keeps Ahmedabad in the database for transparency but excludes it from the main modelling population and most comparisons, then checks sensitivity to its inclusion.

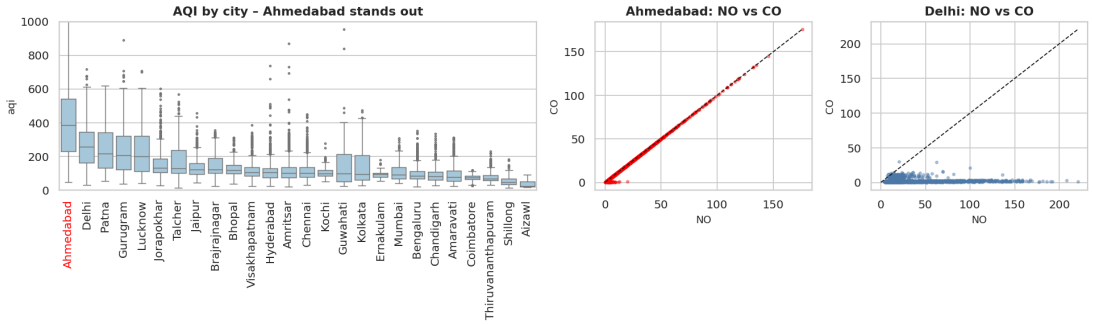


Figure 3.5: Evidence used to flag Ahmedabad as a suspect series. The flag is a data-quality decision, not a claim about the city's true air quality.

Chapter 4

Database design and data pipeline

4.1 Why use a database instead of only a CSV?

A CSV is convenient for exchanging a rectangular table. It is less convenient for maintaining relationships, validating keys, recording transformations and storing analysis outputs. SQLite provides relational tables, keys, constraints, views and SQL queries in a single local database file. The project uses both a database route and a CSV modelling-table route.

4.2 The main relational tables

- `raw_city_day`: imported source records in the database's staging layer.
- `dim_city`: one row per city, with a numeric city identifier and metadata such as state and region.
- `fact_aqi`: one row per city-date, linked to the city dimension using a foreign key. It stores pollutant measurements, AQI and quality flags.
- `festival_calendar`: lookup of the event dates used in the analysis.
- `audit_log`: named pipeline steps and affected-row counts.
- `model_results`: stored evaluation results across modelling tasks.

The project also creates views such as `v_aqi_full` and `v_city_summary`. Indexes support common lookups by date and city/date.

4.3 Keys and integrity

The fact table uses a composite primary key on (city_id, date). This expresses the assumption that there should be at most one daily record for a city. A foreign key ensures that every fact row refers to a city in the dimension table. A primary key prevents duplicate city-date pairs; the foreign-key check tests for orphaned city identifiers. The project checks both and finds no foreign-key violations. The cleaned fact table contains 28,157 rows.

4.4 What the SQL cleaning pipeline does

The SQL pipeline consists of named steps. It trims city names, unifies known aliases, removes duplicate city-date rows, deletes fully empty rows, loads the city dimension and fact table, converts known placeholder zeros to NULL, checks AQI categories and flags suspicious/outlier records. Each operation is recorded in the audit log with its affected-row count.

Audit operation	Rows affected
Trim city names	0
Unify city aliases	0
Delete duplicate rows	0
Delete completely empty rows	1,374
Load city dimension	26
Load fact table	28,157
Convert placeholder zeros to NULL	5,913
Fix AQI bucket	0
Flag outliers	547
Flag suspect-city rows	1,559

Zero affected rows for duplicate removal and name normalization are still useful results: they mean the checks were performed, and the data already satisfied those conditions. Similarly, zero rows changed by the AQI-bucket correction means the stored labels were already consistent with the numeric thresholds.

4.5 What SQL concepts does the project demonstrate?

The 14 analytical queries exercise the important building blocks of relational analysis:

- `SELECT`, filtering and sorting to inspect records;
- `GROUP BY` and aggregate functions for city/month summaries;
- joins between the fact and city dimension tables;
- `CASE` expressions for category and conditional summaries;
- date functions for monthly and yearly aggregation;
- window functions for ranking and time-based comparisons;
- transactions and a demonstrated rollback to show how a change can be undone.

Data-lineage caveat. The repository README describes the database raw table as exactly what the CSV contained. In the executed pipeline, however, the staging table is also modified by the SQL cleaning steps (for example, empty rows are deleted and placeholder zeros are changed to NULL). The original CSV file in `data/raw/` remains the immutable source copy, but `raw_city_day` should be understood as a mutable staging layer rather than a pristine raw archive. A stronger production design would preserve a byte-for-byte immutable source table and clean into a separate staging/curated layer.

4.6 Where the database pipeline lives

The SQL definitions are in `sql/schema.sql`, `sql/clean_pipeline.sql` and `sql/queries.sql`. Reusable orchestration is in `src/db.py` and `src/pipeline.py`. Notebook 01 demonstrates database creation, loading, querying, updating, deleting, transaction rollback and validation.

Chapter 5

Data cleaning and preparation for modelling

5.1 Why cleaning decisions are not cosmetic

A machine-learning algorithm does not know whether a zero is a physically meaningful concentration, a missing sensor reading, a corrupt value or a data-entry convention. It treats the number as data. Cleaning is therefore part of the scientific definition of the problem, not just a formatting step.

5.2 Empty rows and duplicates

The SQL pipeline removes 1,374 records with no meaningful observation. The uniqueness check finds zero duplicate city-date records. Since the fact table is keyed by city and date, this is also enforced structurally. No city alias or whitespace changes were required by the shipped data.

5.3 Placeholder zero values

For CO, NOx, Benzene, Toluene and Xylene, repeated zero values appear in patterns that suggest they may represent unavailable readings rather than genuine zero concentration. In particular, the source shows many identical zero entries for aromatic pollutants in Mumbai. The pipeline changes these known placeholder zeros to NULL (missing) rather than allowing the model to interpret them as real absence of pollution. This is a domain-informed cleaning rule and should be documented because it changes the data distribution.

5.4 Short-gap forward filling

The feature builder first creates a continuous daily calendar for each city. It then forward-fills pollutant measurements for gaps of at most three days, within that city only. Forward filling means carrying the most recently observed value forward:

$$x_t^{\text{filled}} = x_{t-k}, \quad 1 \le k \le 3,$$

when the intervening observations are missing and an earlier value exists. This is causal because it never uses a future measurement. It is limited to short gaps because carrying a reading through a long outage would be unjustified. The target AQI is never filled.

The project reports 275 missing PM2.5 runs; the median gap length is two days, 44% are at most one day, 69% at most three days and 81% at most seven days. A three-day limit therefore fills a substantial fraction of short interruptions without pretending that long outages are known.

5.5 Model-time imputation

After limited forward filling, remaining feature gaps are handled inside model pipelines. Linear and neural-network pipelines use median imputation followed by standardisation. Tree-based models in the shipped scikit-learn implementation can receive missing numeric values directly. Crucially, the imputer and scaler are part of the pipeline and are fitted only on the training fold. They are not fitted once on the full dataset.

The project compared three imputation approaches using time-series validation on training data:

Method	**MAE**	**RMSE**	R^2
Median	34.62	47.39	0.78
Median + missing indicator	34.99	48.26	0.77
MICE / iterative imputation	35.23	46.84	0.78

The results are essentially tied. The simplest median approach is selected because the more elaborate iterative method does not produce a meaningful improvement and adds complexity. It is important not to overstate the comparison: the MAE/RMSE differences are small, and a single result table cannot establish a universally best imputer.

5.6 Outliers versus suspect data

An outlier is an extreme observation; it is not automatically wrong. The project flags 547 outlier rows using unusually high AQI or sensor-ceiling logic but retains them rather than blindly deleting them. High pollution events may be real and are precisely the cases a model should try to understand. Ahmedabad is treated separately as a suspect series based on multiple internal inconsistencies. This distinction is important: flagging supports inspection, while deletion would discard evidence.

5.7 Rows eligible for models

The modelling table marks a same-day row valid only when AQI is known, at least one particulate measurement is available, and at least three modelled pollutant measurements are present. This mirrors the broad completeness logic used in the CPCB method. A next-day row must additionally have a known AQI on the following calendar day. After excluding the suspect Ahmedabad series, 23,500 observations qualify for same-day modelling and 23,030 for next-day validity before the final feature/target-date split.

Chapter 6

Feature engineering

6.1 What is a feature?

A feature is an input variable supplied to a model. The target is the quantity the model is asked to predict. In same-day regression the target is numeric AQI. In same-day classification the target is the AQI bucket. In next-day forecasting the target is tomorrow's AQI or tomorrow's bucket.

A raw date string is not directly useful to most algorithms. Feature engineering transforms it into interpretable variables such as month, season, weekend and event flags. For forecasting, it also creates history-based variables such as yesterday's AQI and a rolling average.

6.2 Calendar features

The project extracts year, month, day, weekday, day of year and quarter. It also creates weekend, season and winter indicators. The numerical month is represented cyclically:

$$m_{\sin} = \sin\left(\frac{2\pi m}{12}\right), \tag{6.1}$$

$$m_{\cos} = \cos\left(\frac{2\pi m}{12}\right). \tag{6.2}$$

This encoding avoids treating December and January as far apart: they are neighbours in the annual cycle. Seasons are assigned as Winter (December–February), Summer (March–May), Monsoon (June–September) and Post-monsoon (October–November). The calendar year is deliberately not used as a model input, because a tree model can memorise historical year levels but does not naturally extrapolate to a new year.

6.3 Festival and event features

The feature builder marks windows around Diwali, Holi and Dussehra, as well as the north Indian stubble-burning season (15 October–30 November) and the 2020 lockdown period (25 March–31 May). It also stores signed distance to the nearest festival date. A flag answers whether a date lies within a defined event window; a distance feature captures how close the date is to the central event.

These dates are hard-coded in `src/config.py`. They should be double-checked against official calendars before reusing the pipeline for a new analysis. The Diwali window is asymmetric (one day before through three days after), while Holi and Dussehra use a one-day-before/one-day-after window.

6.4 Lag and rolling features for forecasting

For every city, the project creates AQI lags at 1, 2, 3 and 7 days, rolling mean AQI over 3 and 7 days, rolling 7-day standard deviation, lagged PM2.5 and PM10, and one-day AQI change. A lag is simply an earlier value aligned with the current prediction date. For example:

$$\text{AQI lag 1 at date } t = \text{AQI}_{t-1}.$$

A rolling mean smooths short-term noise and represents recent level:

$$\overline{\text{AQI}}_{7,t} = \frac{1}{7}\sum_{j=0}^{6} \text{AQI}_{t-j},$$

when enough observations are available. The next-day target is created as `aqi_next = AQI shifted by -1 within each city`.

A lag must be a true previous calendar day, not merely the previous row in a sparse table. The code re-inserts missing dates for each city before making lags and checks that each city calendar is contiguous. Otherwise, a 1-row shift could accidentally mean three days earlier after a data gap.

6.5 The 23 same-day candidate features

There are 23 candidate inputs: eleven pollutant variables, five calendar variables (cyclic month pair, weekday, weekend, winter), five event variables (Diwali, Holi, Dussehra, stubble season, lockdown), and two categorical variables (city and season). After preprocessing, categorical city and season values are one-hot encoded, so the number of numeric model columns becomes larger than the original feature count. For the full feature set the pipeline has 42 transformed columns.

Chapter 7

Exploratory data analysis: what does the dataset say?

7.1 AQI distribution and category balance

EDA is the stage where we learn what kind of data we have before choosing a model. A distribution plot shows where AQI values are concentrated, whether there is skew and whether extreme values exist. The full-history bucket proportions are approximately: Good 5.7%, Satisfactory 34.8%, Moderate 36.7%, Poor 10.8%, Very Poor 9.0% and Severe 3.0%. Thus most observations are in the Satisfactory or Moderate bands, while the extreme classes are relatively rare.

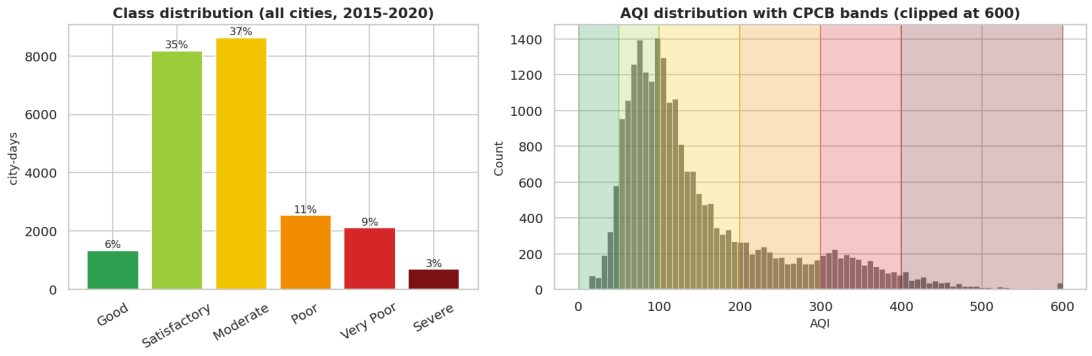


Figure 7.1: Distribution of AQI over the analysis population. The long upper tail indicates occasional severe pollution episodes.

7.2 City-to-city differences

Across the longer historical record, the highest mean AQI values are observed for Delhi (about 260), Patna (about 241) and Gurugram (about 225). Lower averages include Coimbatore (about 73), Thiruvananthapuram (about 76) and Bengaluru (about 94). These are descriptive averages over uneven city coverage and should not be interpreted as a perfectly balanced ranking of cities.

7.3 Trends over years and months

The yearly profile suggests lower observed AQI in later years, but this should not be interpreted as a clean long-term improvement without considering sensor coverage, which cities contribute observations, missingness and the COVID-19 period. Monthly profiles show a strong seasonal signature. Northern cities peak around November, with a peak-to-trough ratio of roughly three, whereas southern cities vary by about 1.6 times.

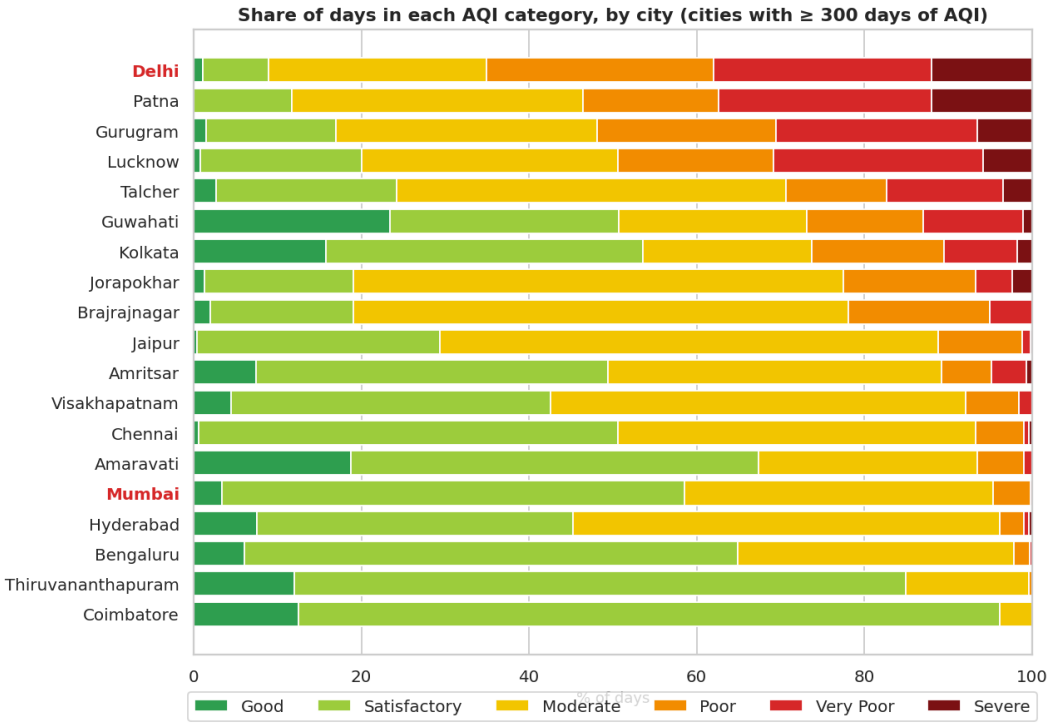


Figure 7.2: City-level AQI category composition. This makes it possible to compare the frequency of poor-or-worse days, not only average AQI.

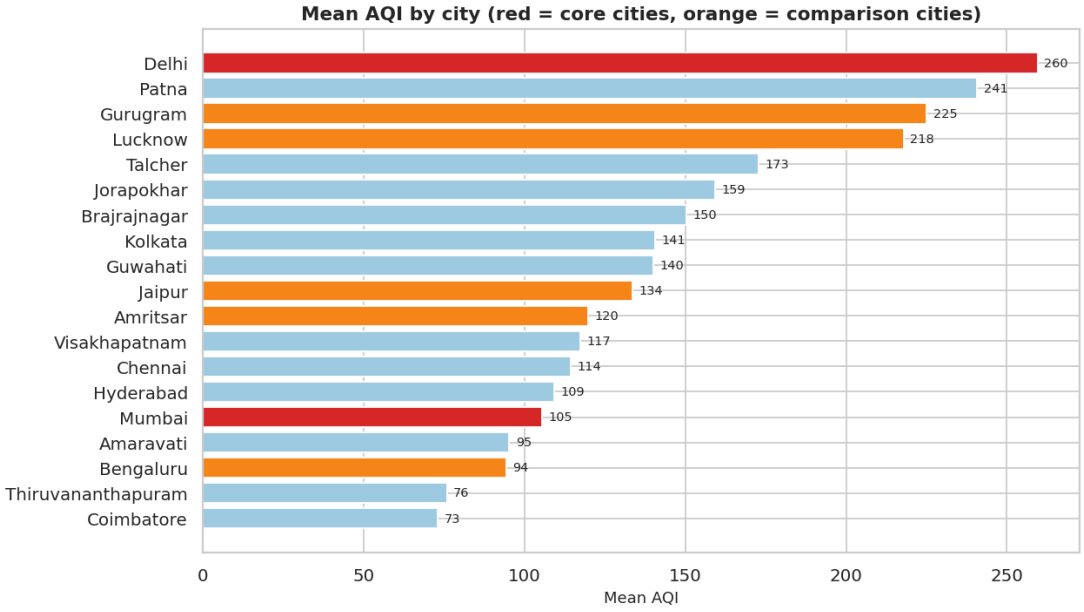


Figure 7.3: Mean AQI by city over the available historical record. Unequal coverage and Ahmedabad's suspect series affect interpretation; Ahmedabad is flagged separately.

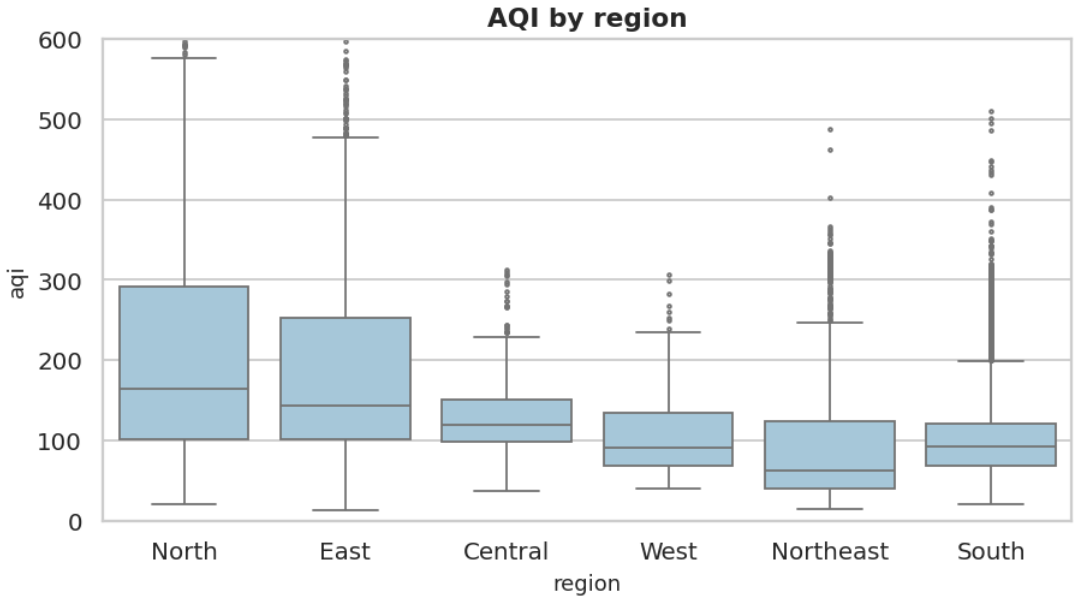


Figure 7.4: Regional summary used to describe broad geographic differences in the historical sample.

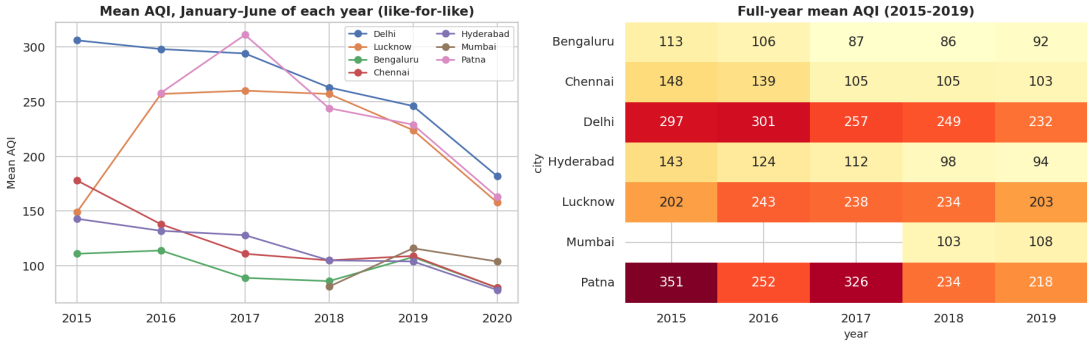


Figure 7.5: Yearly AQI trend. Changes in the composition and coverage of cities must be considered before making causal or policy claims.

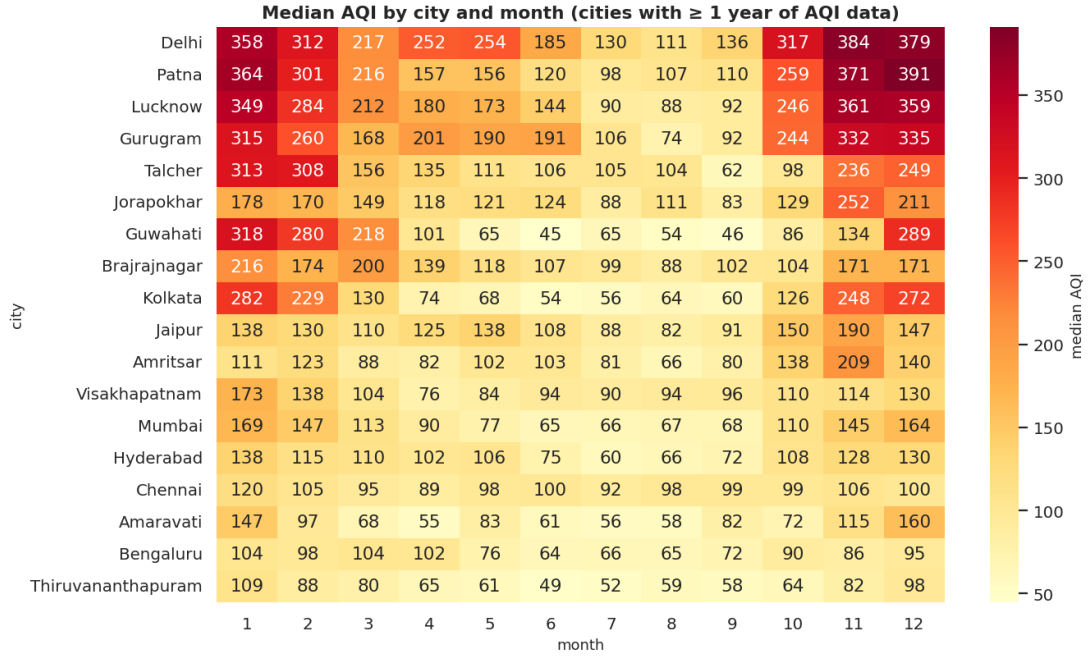


Figure 7.6: Month-by-city structure reveals different seasonal cycles across the country.

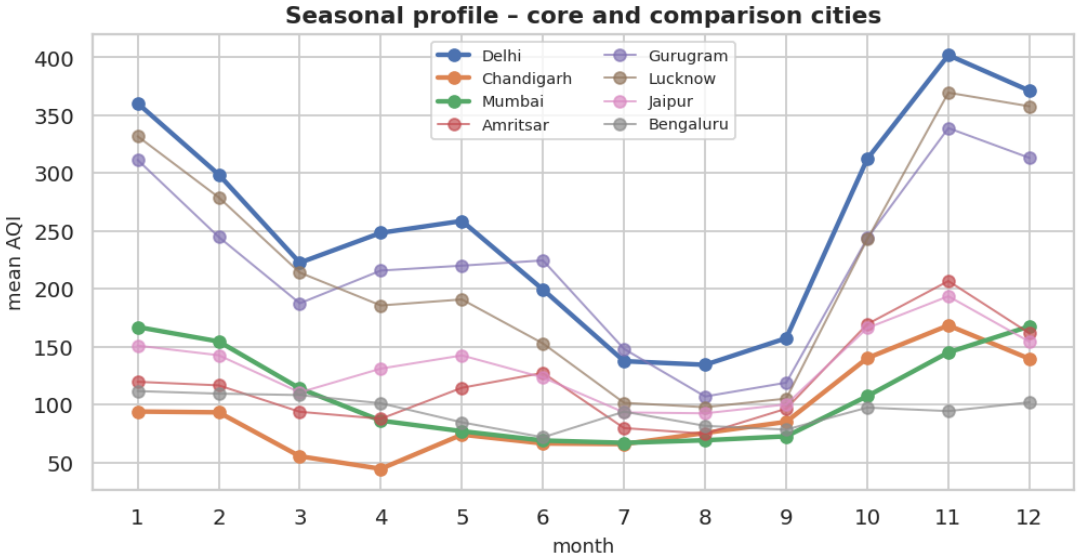


Figure 7.7: Seasonal AQI profile for core cities. Winter and post-monsoon periods have higher average AQI than the monsoon period.

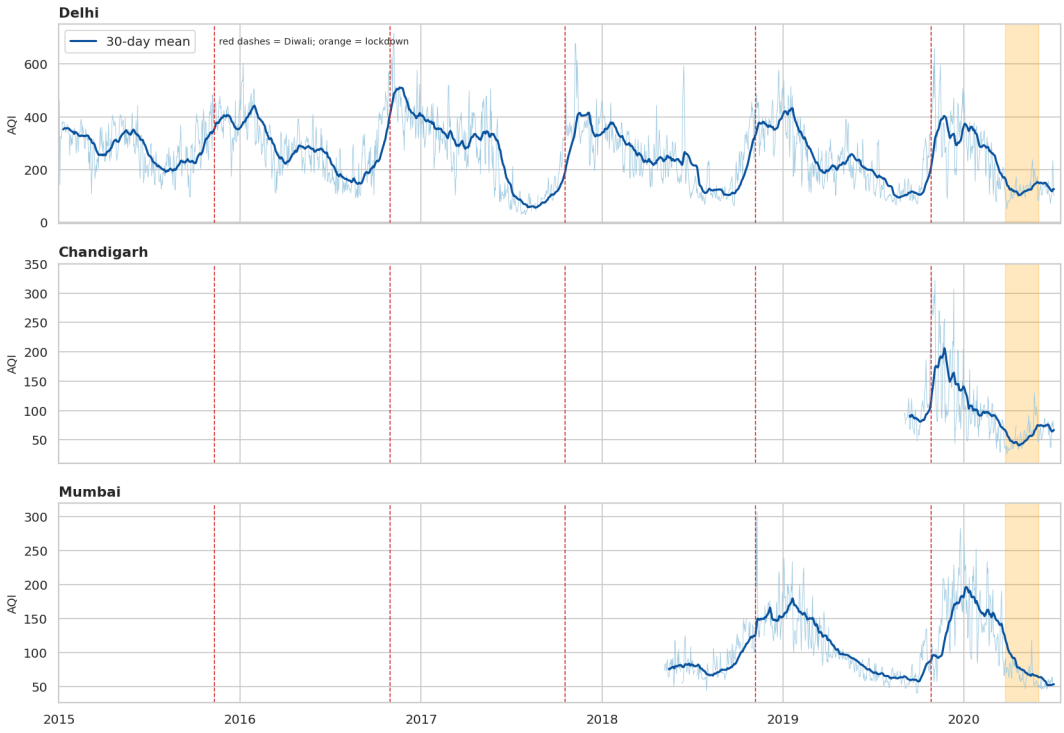


Figure 7.8: Time series of the core cities, showing both long-run variation and abrupt pollution episodes.

The mean seasonal AQI in the analysis is approximately 204 in Winter, 191.4 in Post-monsoon, 134.5 in Summer and 100.3 during Monsoon. Rainfall and atmospheric mixing are plausible mechanisms for lower monsoon concentrations, while winter stagnation and regional burning may contribute to higher northern values. However, the dataset itself has no weather variables, so this is interpretation rather than a causal test.

7.4 Weekday effect

The weekend-minus-weekday difference is about -1.64 AQI points, with a reported p-value near 0.051. The largest average weekday swing is only about 4.6 points. On this dataset, the weekday pattern is small compared with the city and seasonal differences. A p-value close to 0.05 should not be described as proof of a practically important weekend effect.

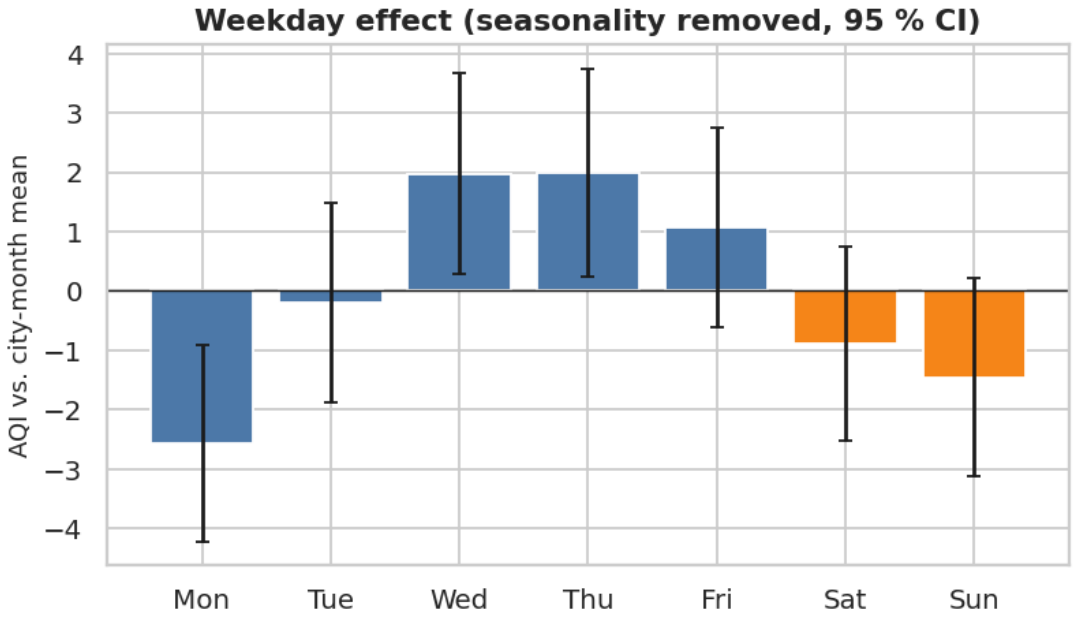


Figure 7.9: Weekday profile. The average differences are small compared with the scale of city and seasonal variation.

7.5 Pollutant relationships

Spearman correlation measures monotonic association and is less sensitive than Pearson correlation to a few extreme observations. PM2.5 and PM10 both have Spearman correlation around 0.88 with AQI in the full analysis population. CO is around 0.51; NOx, NO and NO2 are around 0.46, 0.45 and 0.43. Lower associations are observed for NH3, Toluene, SO2, O3 and Benzene. Correlation does not establish causation, and it does not mean one pollutant is unimportant when correlated with another.

7.6 Festival, stubble-burning and lockdown analyses

The project goes beyond plotting time series by comparing event windows with placebo dates. A real event window is compared with dates offset from the event, intended to represent the seasonal background without the event itself. For Diwali, the baseline is days -10 through -4 and the event window is days -1 through +3. The placebo analysis is important because Diwali occurs during a period when AQI may already be rising seasonally.

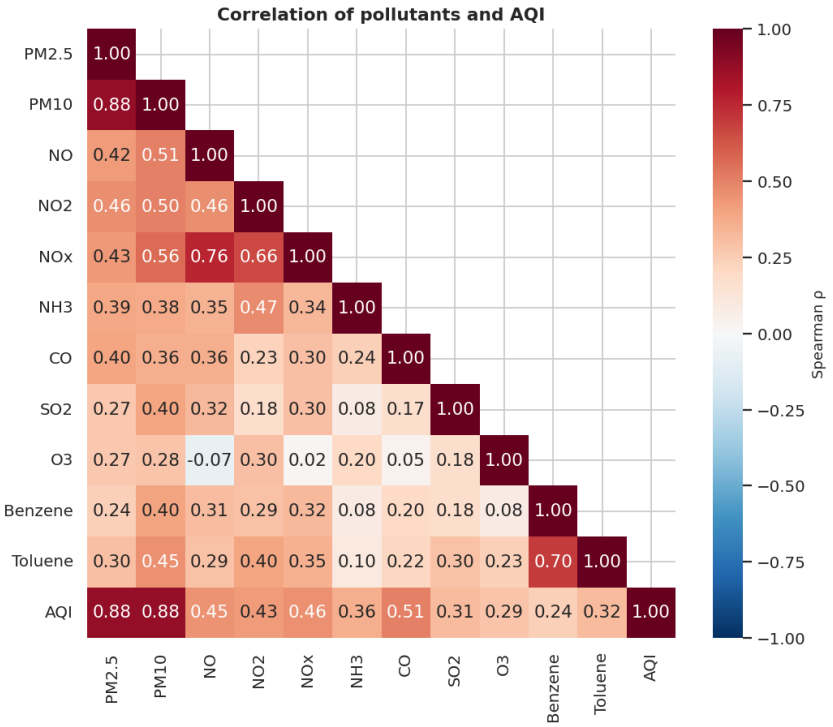


Figure 7.10: Spearman correlation matrix for pollutant and AQI variables. Strong association between PM2.5, PM10 and AQI is expected because particulate sub-indices often determine the final AQI.

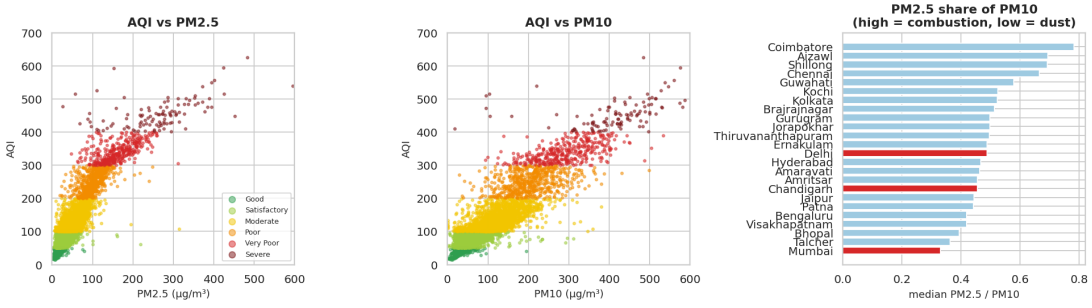


Figure 7.11: Relationship between particulate concentrations and the reported AQI. Nonlinear and segmented structure is expected from the AQI breakpoint formula.

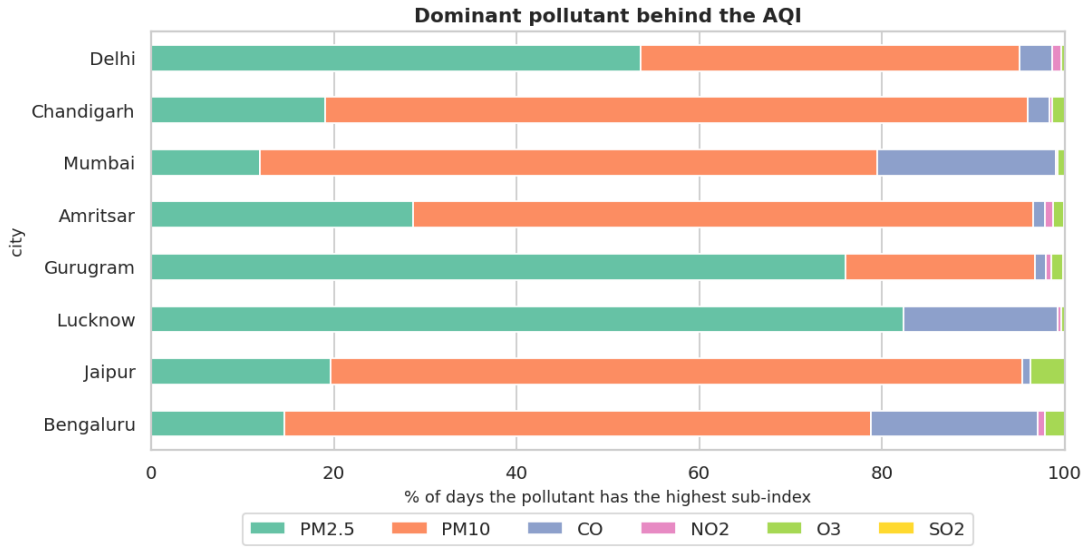


Figure 7.12: Dominant pollutant according to the reconstructed sub-index calculation. PM2.5 and PM10 dominate many city-days.

For all cities, AQI rises by about 37.8% in the Diwali window compared with the pre-event baseline, while placebo dates rise by about 4.8%. The excess is approximately 33 AQI points. In North India, the real increase is 46.3% versus -5.7% for placebo, an excess of around 52 points. The analysis reports very small permutation-style probabilities for the real effect exceeding placebo. Holi shows no comparable excess. Dussehra is elevated, but it is especially confounded with the autumn transition and should be treated as an upper-bound association rather than a clean event effect.

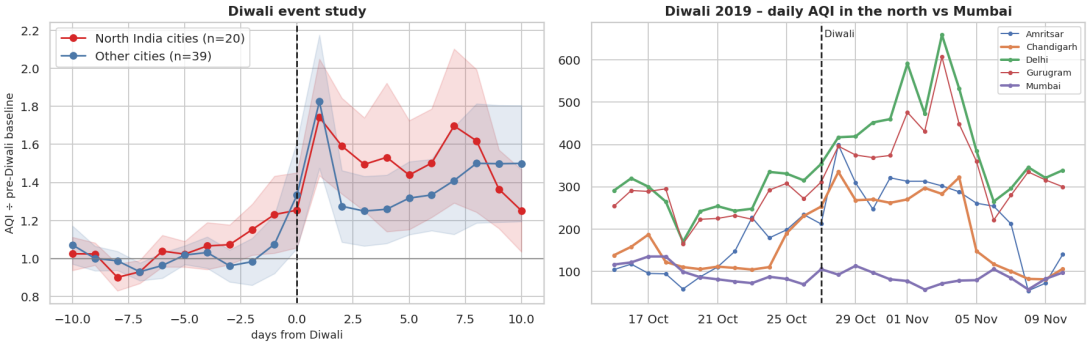


Figure 7.13: Diwali event study compares the observed event window with placebo windows to account for seasonal drift.

The mean AQI in the stubble-season window is about 125 in northern cities and 94 elsewhere, with a reported p-value near 0.00019. This is consistent with a regional seasonal pollution burden but does not isolate the effect of crop-residue burning from weather, traffic or other co-occurring sources.

During the 2020 COVID-19 lockdown, the median AQI change across 18 cities is about -31% compared with the same dates in the preceding year. PM2.5 falls around 37%, PM10 40%, NO2 36%, CO 27%, SO2 17% and O3 changes by only about -2%. Delhi's AQI falls around 44% and Mumbai's around 27%. This is an observational before/after comparison; weather, seasonal drift and other changes are not fully controlled, so it should not be described as a definitive causal estimate of lockdown impact.

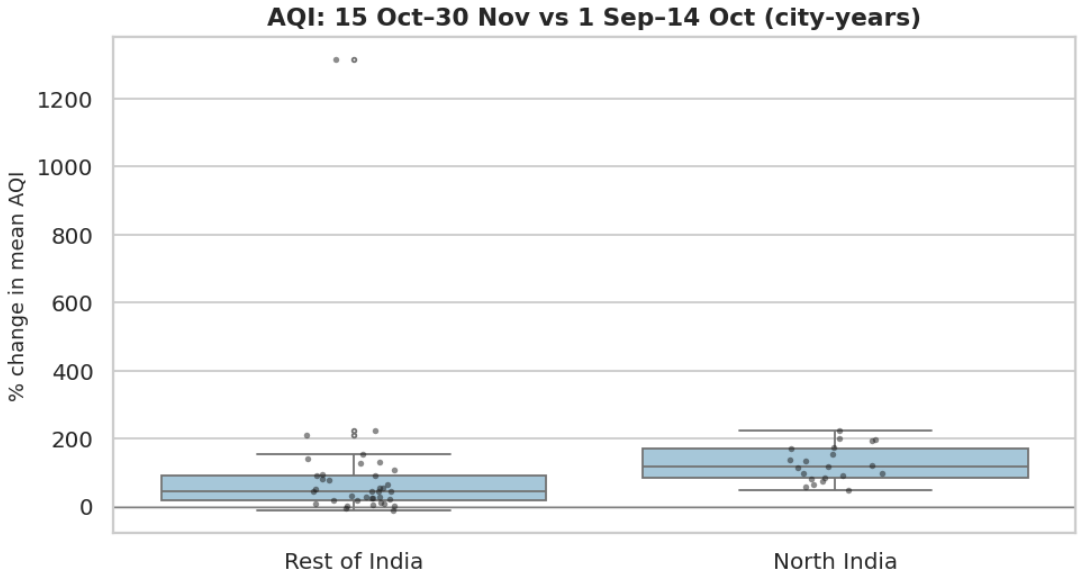


Figure 7.14: AQI during the defined stubble-burning season in North India versus other regions.

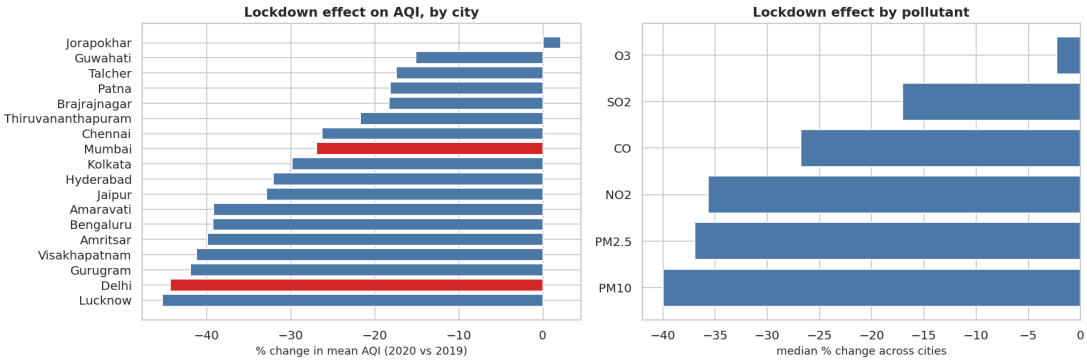


Figure 7.15: Pollutant and AQI changes observed during the lockdown comparison. This is descriptive evidence, not a fully controlled causal experiment.

7.7 What EDA contributes to modelling

EDA motivates several later decisions: Xylene is removed because it is highly incomplete; particulate matter deserves attention because it is strongly associated with AQI; city and season need careful handling because cities are unevenly observed; class imbalance motivates macro-averaged metrics and class weights; and chronology motivates time-based rather than random evaluation. EDA is not decoration: it is how the modelling assumptions are made visible.

Chapter 8

Feature reduction: finding a smaller useful input set

8.1 Why reduce features?

A model can use many inputs, but more inputs do not always improve prediction. Redundant or weak features may increase noise, complexity and overfitting. Feature reduction asks whether a smaller set retains predictive information. It can also make a model easier to explain and cheaper to collect in a future system.

8.2 Methods used

1. **Correlation and mutual information:** correlation measures association; mutual information can detect nonlinear dependence.
2. **Variance inflation factor (VIF):** estimates how strongly a numeric predictor can be explained by other predictors. Large VIF can indicate multicollinearity in linear models.
3. **Correlation filter:** removes one variable from pairs that exceed a chosen correlation threshold, while also considering target relevance.
4. **Permutation importance:** randomly shuffles one feature and measures how much model performance degrades. A large increase in MAE means the model relied on that feature.
5. **PCA:** combines correlated original variables into orthogonal principal components that preserve directions of high input variance.

8.3 Why the naive correlation filter underperforms

PM2.5 and PM10 are strongly correlated (roughly 0.81 in the feature-screening subset), so a simple filter drops one of them. Benzene and Toluene are also strongly correlated (about 0.76). But AQI is the maximum of pollutant-specific sub-indices, not an average. PM10 can be the pollutant that sets AQI on some days while PM2.5 sets it on others. A feature can therefore be correlated with another feature and still provide unique information about the maximum.

The correlation-filtered feature set retains 21 candidates but its Gradient Boosting test performance falls to $R^2 = 0.874$ and MAE 19.07, compared with $R^2 = 0.908$ and MAE 16.46 for the full feature set. The lesson is that pairwise correlation alone is not a sufficient criterion for dropping variables.

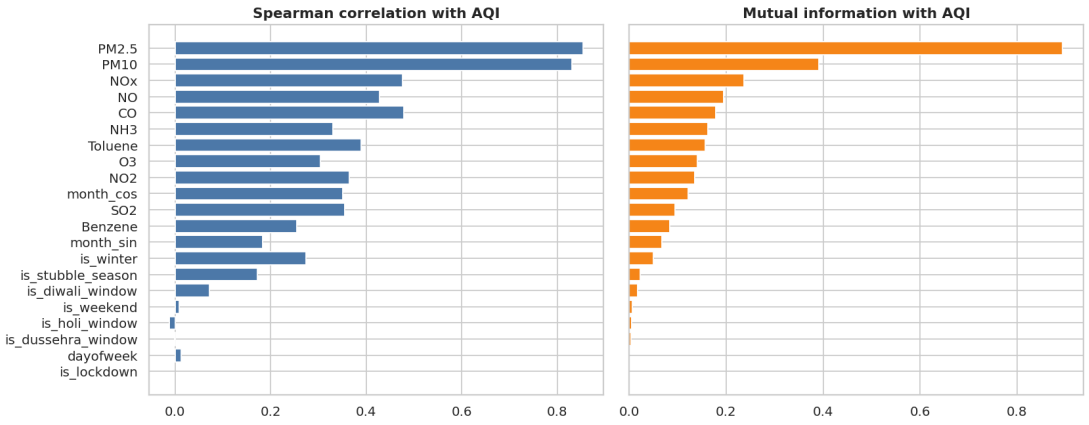


Figure 8.1: Univariate feature relevance measured using target association and mutual information. These scores are screening tools, not proof of causal importance.

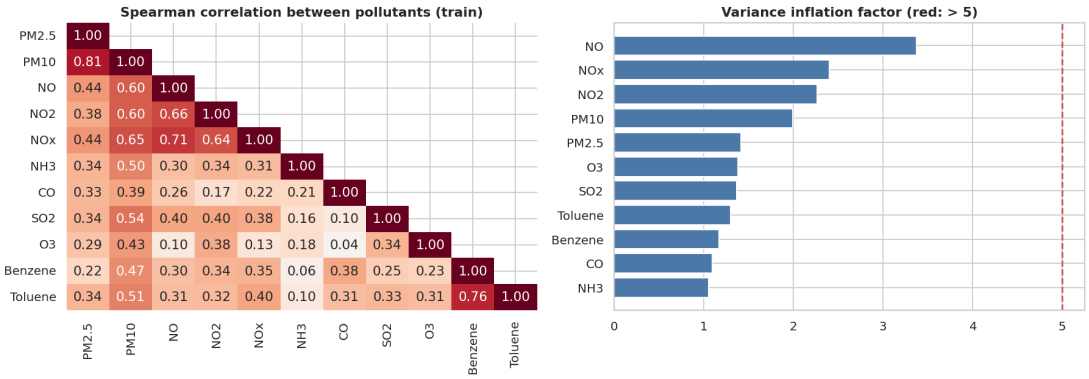


Figure 8.2: Correlation structure and VIF diagnostics. VIF values remain below about 3.4, so severe linear multicollinearity is not apparent.

8.4 Permutation importance and the four-feature set

The permutation analysis shows the largest MAE increase when PM2.5 is shuffled (about +41.85 points), followed by PM10 (+12.99). CO (+1.83), O3 (+1.63) and city (around +1.59) have smaller effects; remaining features are below roughly 1.5 points. An expanding-window four-fold time-series CV was used to select a compact set within a tolerance of 0.005 in R^2. The selected features are PM2.5, PM10, CO and O3.

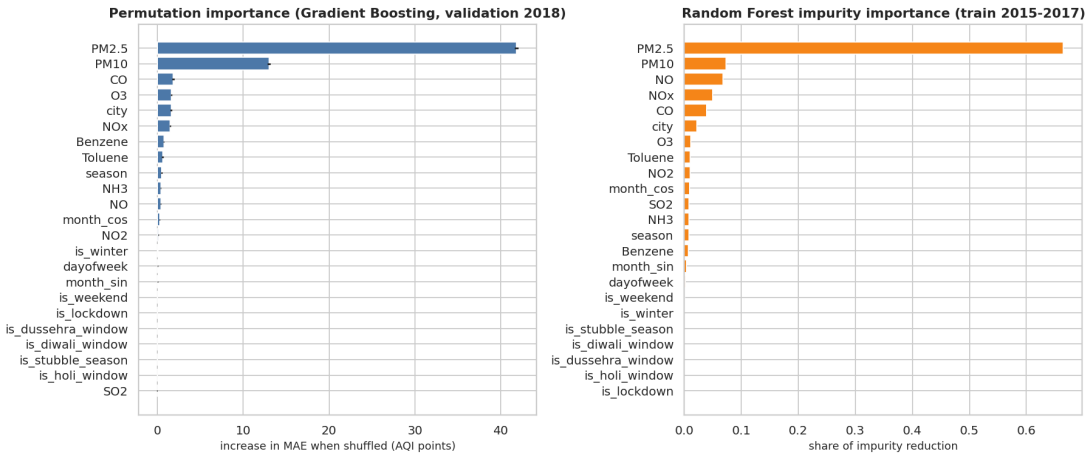


Figure 8.3: Permutation importance: the change in error after a feature is shuffled. The effect is model- and dataset-dependent, and correlated inputs can share importance.

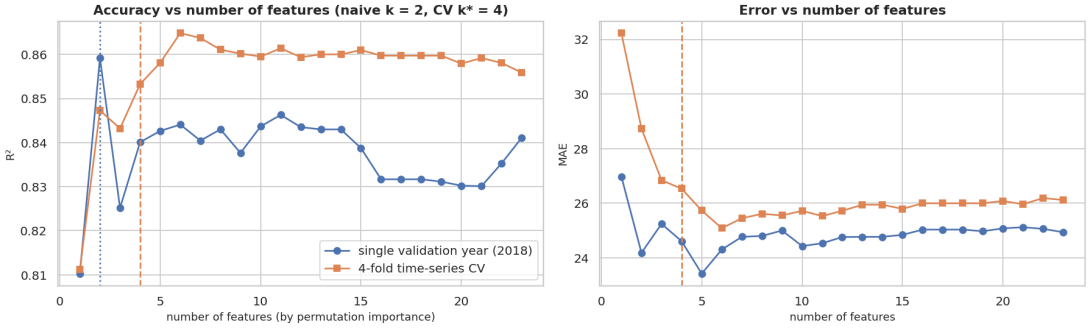


Figure 8.4: Feature-count curve used to assess how much performance is retained as features are added.

The final four-feature set has test $R^2 = 0.905$ and MAE 16.19 for Gradient Boosting, compared with full-set $R^2 = 0.908$ and MAE 16.46. It therefore reduces the number of original predictors from 23 to four with essentially unchanged test performance in this experiment.

8.5 PCA: what it does and why it did not win here

PCA first centres and scales the input variables, then finds directions (principal components) that explain the largest variance in the input data. The first component explains the largest fraction, the second explains the largest remaining fraction, and so on. Keeping enough components to preserve 95% of pollutant variance required ten principal components in this dataset. The PCA feature set then adds twelve non-pollutant features for a total of 22 inputs.

The PCA + Gradient Boosting test result is $R^2 = 0.823$ and MAE 21.00, worse than both the full feature model and the selected four-feature model. This is not a contradiction: PCA preserves variance in the inputs, not necessarily information that is useful for predicting AQI. It also creates mixtures of pollutants that are harder to interpret and may obscure the max-of-sub-indices structure.

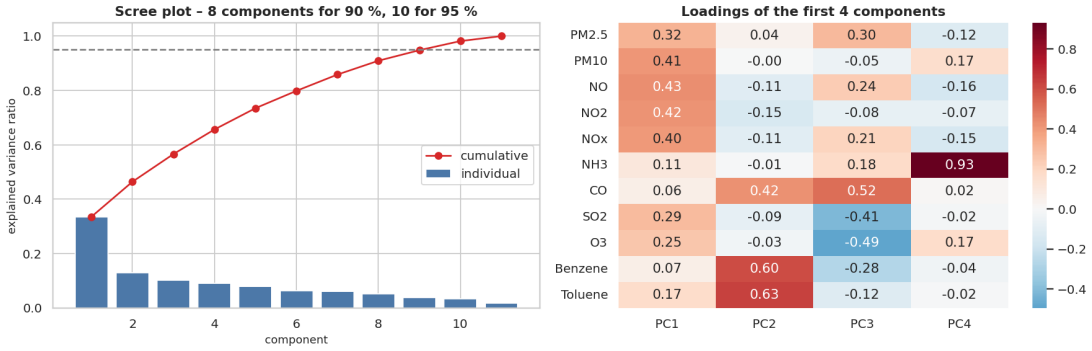


Figure 8.5: Explained variance and component structure for pollutant PCA. Ten components were needed to retain 95% of pollutant variance.

8.6 A validation lesson

A naive single-year validation pick selected only PM2.5 and PM10 and then achieved a worse test score ($R^2 = 0.883$, MAE 19.53). The final four-feature selection used expanding-window CV and performed better. This is an important methodological lesson: feature selection is itself model selection, so it must be done inside the training period and across multiple temporal validation folds. The project transparently preserves both results rather than hiding the weaker preliminary choice.

Feature set (HistGB)	**Features**	**Test R^2**	**Test MAE**
Full set	23	0.908	16.46
Importance-selected	4	0.905	16.19
Correlation filter	21	0.874	19.07
PCA + remaining variables	22	0.823	21.00
Naive top-2 pick	2	0.883	19.53

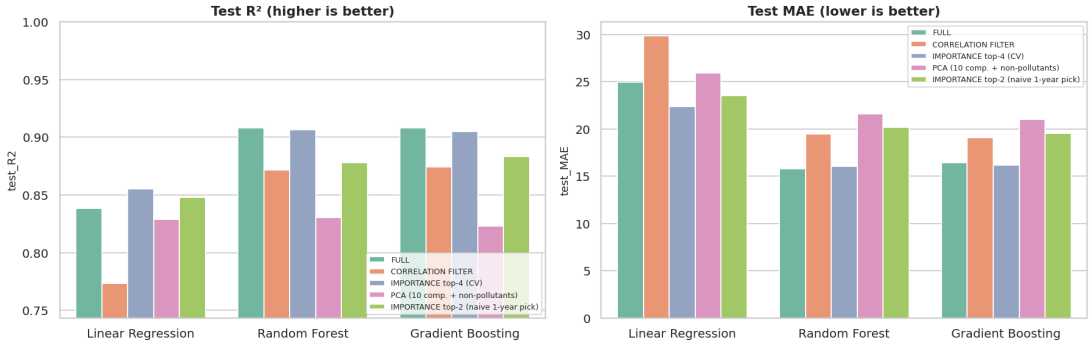


Figure 8.6: Direct comparison of the full, correlation-filtered, PCA, two-feature and four-feature alternatives.

Chapter 9

Machine-learning evaluation: the rules of a fair test

9.1 Why random train/test splitting is inappropriate

For ordinary independent samples, random splitting can be reasonable. For time series, it can put observations from later dates in training and earlier dates in testing. This leaks future regimes into the model-development process and makes the test set less representative of a future deployment.

The project uses a chronological split: dates before 1 January 2019 form the training period; dates from 1 January 2019 to 1 July 2020 form the held-out test period. In same-day modelling this gives 12,541 training observations and 10,959 test observations after validity and suspect-city filtering. Chandigarh's observations occur only in the test period, so it acts as a useful unseen-city transfer check.

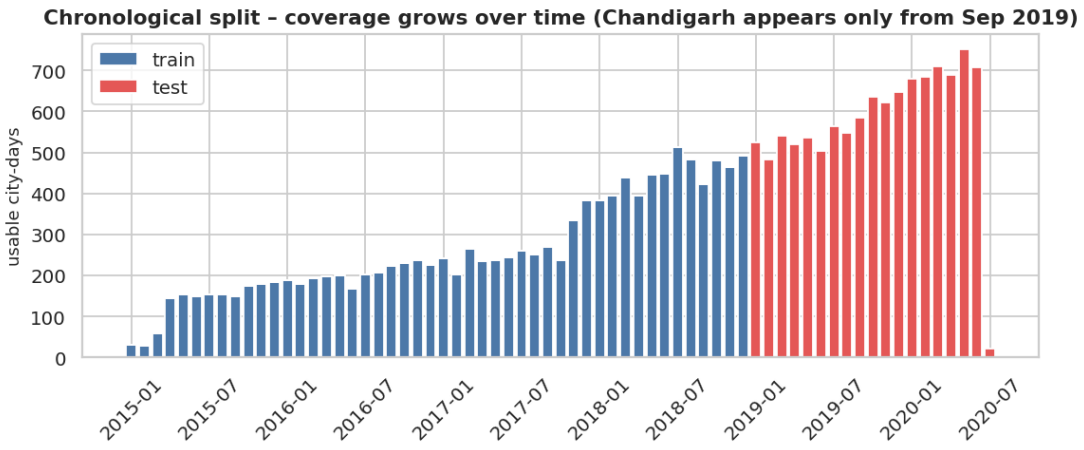


Figure 9.1: Chronological train/validation/test logic. The 2019–2020 period is held out from model fitting and tuning.

9.2 Validation inside training

Hyperparameters are settings that control model complexity, such as tree depth or the number of leaves. They are selected using randomized search over a defined parameter space and time-series cross-validation. In a time-series split, each validation fold occurs after the training portion. The main metric used for regression tuning is negative MAE; classification tuning uses macro-F1.

The search uses a fixed random seed (42) for reproducibility and a small randomized set of candidate configurations. The test period is not used for parameter search. The fitted preprocessing steps

(imputation, scaling, encoding and, where used, PCA) live inside scikit-learn pipelines so they are fitted separately within each training fold.

9.3 A caveat about panel time-series validation

The data contain multiple cities observed on the same dates. The implementation sorts records by date before applying ordinary `TimeSeriesSplit`. This is time-ordered, but a stricter panel-data protocol would split on unique dates and keep every city's observations from a particular date entirely within one side of a fold boundary. That improvement would eliminate the possibility that different cities from the same date are divided across a boundary. The final held-out test split remains chronological.

9.4 Regression metrics

Mean Absolute Error (MAE) is the average absolute difference between the predicted and actual AQI:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|.$$

It is in AQI points, so it is easy to interpret. **Root Mean Squared Error (RMSE)** penalises large errors more strongly:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}.$$

Coefficient of determination compares the model with predicting the test-set mean:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

An R^2 of 1 is perfect; 0 is equivalent to predicting the mean in squared-error terms; a negative value is worse than that baseline.

9.5 Classification metrics

Accuracy is the fraction of correct labels. Precision measures how often a predicted class is correct; recall measures how many actual members of a class are found. F1 combines precision and recall. Macro-F1 calculates F1 for each class and then takes an unweighted average, giving rare Severe observations the same class-level weight as common Moderate observations. Balanced accuracy averages recall across classes. A confusion matrix shows actual classes against predicted classes.

Because the six AQI categories are ordered, the project also reports whether errors are within one adjacent category. Predicting Poor instead of Very Poor is a smaller mistake than predicting Good instead of Severe, even though ordinary accuracy counts both as incorrect.

9.6 Baselines and sanity checks

A baseline gives a minimum standard the model must beat. Same-day regression includes a mean predictor and Linear Regression. Same-day classification includes a majority-class predictor and Logistic Regression. Next-day forecasting includes persistence: predict tomorrow's AQI as today's AQI. Persistence is particularly important because pollution often changes gradually.

The repository also includes shuffled-target sanity checks. A model trained on randomly permuted target labels should not preserve genuine predictive skill. The same-day regression check produces a

negative R^2 (about -0.28), consistent with the expectation that the pipeline is not simply exploiting an accidental label relationship.

Chapter 10

Same-day AQI regression

10.1 Task definition

The regression target is the numeric AQI for the same date. The candidate inputs are the eleven pollutant concentrations plus calendar, event and city/season variables. The compact four-feature analysis uses PM2.5, PM10, CO and O3. The target is continuous, so the model can predict values such as 137.4 or 268.1 rather than selecting a category.

10.2 Models compared

- **Mean dummy regressor:** predicts the mean AQI from the training data; this is a deliberately weak reference.
- **Linear Regression:** fits a weighted sum of features. It is interpretable and provides a simple baseline, but it cannot automatically represent complex nonlinear interactions.
- **Decision Tree:** partitions feature space into rule-like regions and predicts a value within each terminal leaf. It learns nonlinear thresholds but can overfit.
- **Random Forest:** trains many trees on bootstrap/subsamples and averages their predictions, reducing variance compared with a single tree.
- **Histogram Gradient Boosting (HistGB):** builds trees sequentially, with each new tree correcting errors made by the current ensemble. It can capture nonlinear relationships efficiently.
- **MLP neural network:** uses two hidden layers (64 and 32 units in the catalogue) to learn nonlinear combinations. The target is scaled during training and converted back to AQI units for evaluation.

10.3 Held-out test results

Model	MAE	RMSE	R^2
Mean dummy	82.89	95.15	-0.289
Linear Regression	22.37	31.89	0.855
Decision Tree	17.90	28.12	0.887
Random Forest	**15.96**	25.73	0.906
HistGradientBoosting	16.05	25.41	0.908
MLP neural network	16.34	**24.73**	**0.913**

The Random Forest has the lowest test MAE, while the MLP has the lowest RMSE and highest R^2. HistGB is close to both. Model selection based on training-only time-series CV favours the MLP by

CV MAE (24.95, versus 25.39 for the Random Forest). The differences among these three are small enough that it is better to say they are competitive than to claim one is the universal winner.

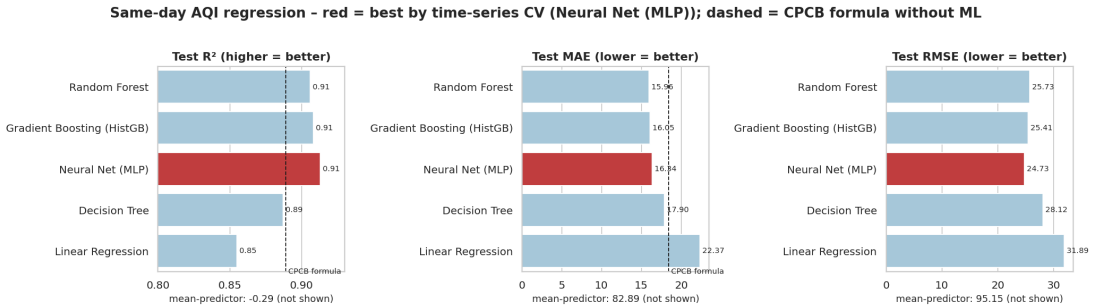


Figure 10.1: Same-day regression comparison across the held-out chronological test period.

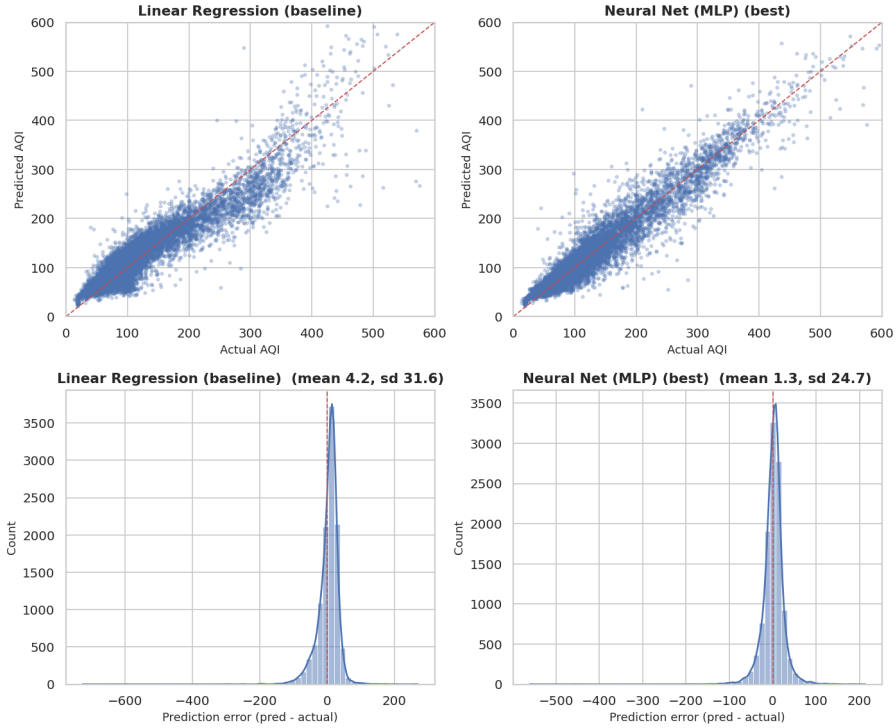


Figure 10.2: Actual-versus-predicted and residual diagnostics. Residual structure indicates where the model still misses patterns.

10.4 How to interpret an MAE of 16

An MAE of about 16 means that the average absolute prediction error is around 16 AQI points on this test sample. It does not mean every prediction is within 16 points, and it does not mean the model is always correct about the AQI category. RMSE is larger (around 25) because a smaller number of larger errors contribute disproportionately when squared.

10.5 Error depends on AQI level

The error increases at the severe end of the distribution. Linear Regression has reported MAE around 17.7 for AQI 0–100, 19.6 for 101–200, 34.5 for 201–300, 50.1 for 301–400 and 95.2 above 400. The nonlinear models reduce the high-end error; for the MLP, the corresponding figures are approximately

10.8, 18.8, 29.4, 23.7 and 45. This indicates that rare extremes are more difficult than common moderate values.

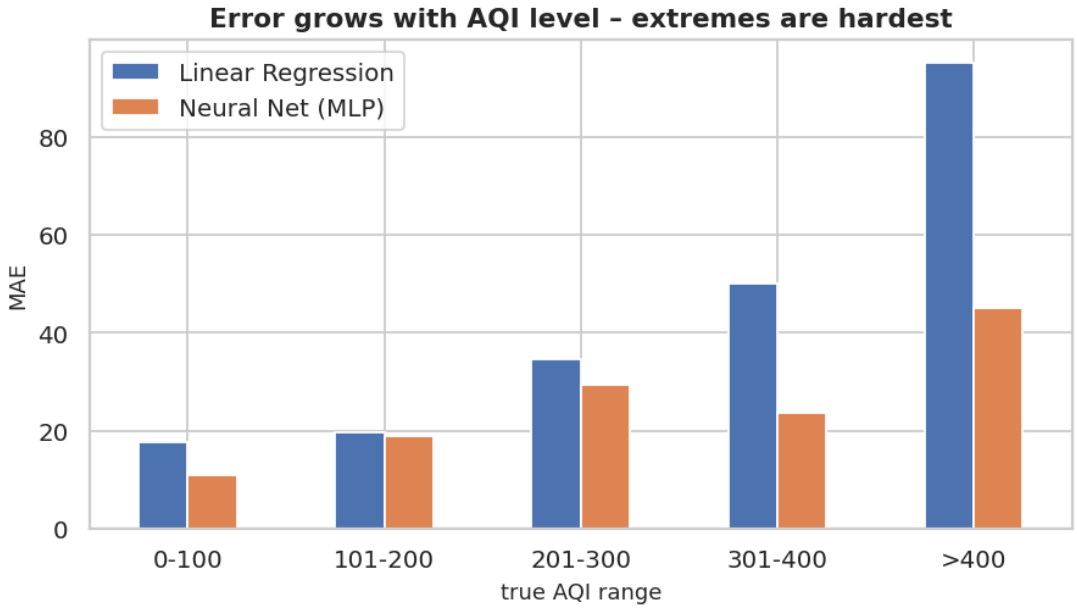


Figure 10.3: MAE by AQI range. Extreme AQI values are less frequent and are more difficult to estimate accurately.

10.6 The unseen-city test

Chandigarh has only 304 observations beginning in September 2019, so it does not appear in the 2015–2018 training period. Yet the MLP's test MAE for Chandigarh is about 14.8. This is a useful transfer result: the model can apply pollutant-to-AQI relationships learned from other cities to a city that was not part of training. It does not prove robust nationwide generalisation because one city with limited coverage is not a substitute for testing on many unseen cities.

10.7 CPCB formula as a non-ML reference

The project re-implements the CPCB sub-index procedure to validate the reported AQI. In the full eligible analysis population, the reconstructed AQI reaches approximately $R^2 = 0.866$ with MAE around 22.6 and agrees on the broad category roughly 72% of the time. In the separate held-out test comparison, the CPCB formula yields MAE 18.40, RMSE 27.93 and $R^2 = 0.889$. These values refer to different evaluation subsets/protocols and should not be mixed as though they were a single score.

The ML models outperform this implementation on the held-out test set, likely because they learn the dataset's own calibration and idiosyncrasies. The comparison is not proof that ML has discovered a better health index; it is a comparison with a reconstruction using daily city-level inputs and a particular breakpoint implementation. The source data do not guarantee that all official averaging windows and completeness rules are available in exactly the same form.

10.8 Partial dependence and interpretation

Partial dependence plots show how the model prediction changes as one feature varies while averaging over the remaining sample. They help inspect learned trends but should not be treated as causal

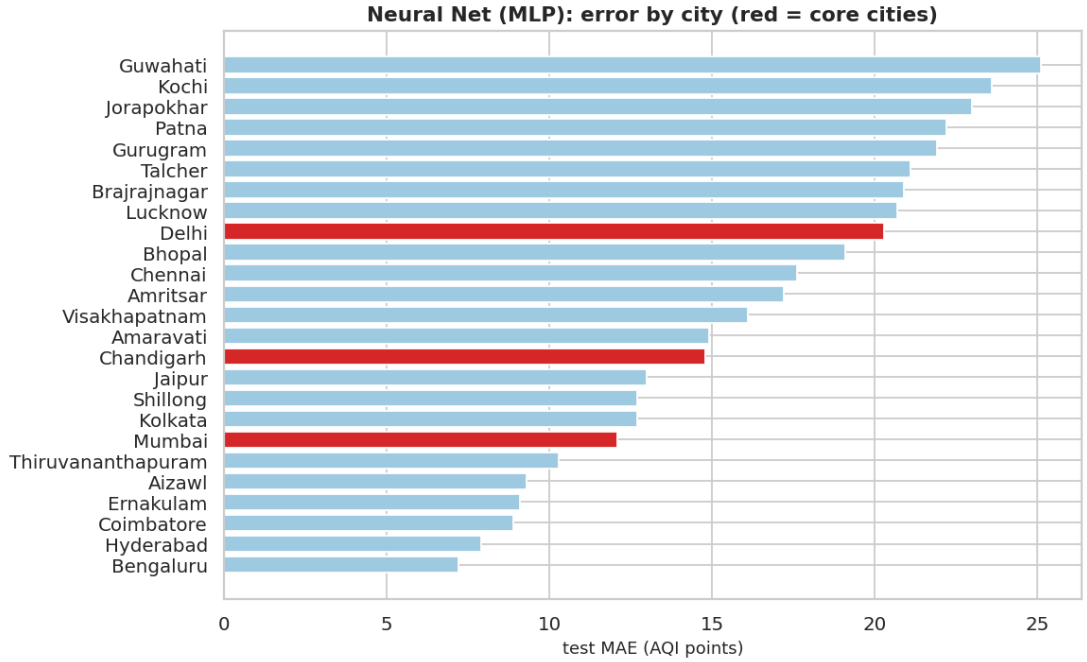


Figure 10.4: Model error by city. City-level performance varies, reflecting data quantity, city-specific regimes and sensor quality.

effects. Correlated pollutant variables can make these plots misleading if an artificial combination of inputs is examined.

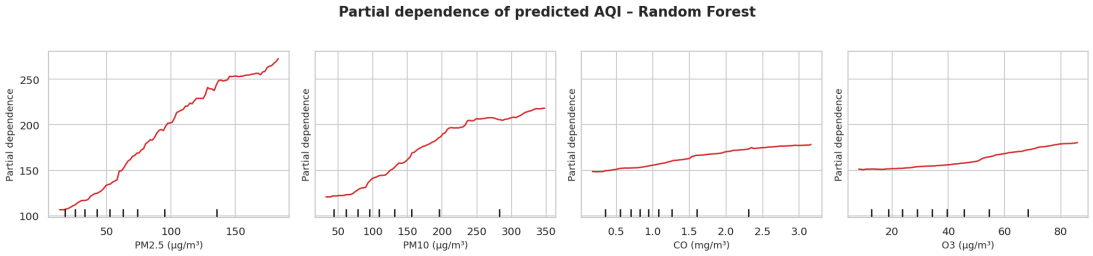


Figure 10.5: Partial dependence curves for selected predictors. These are model-behaviour diagnostics, not causal response curves.

Chapter 11

Same-day AQI classification

11.1 What changes relative to regression?

Classification predicts one of six labels rather than a numeric AQI. The target is obtained by mapping each numeric AQI to its official category thresholds. The features remain pollutant and calendar/city variables. The model is not allowed to use the numeric AQI itself as an input; doing so would leak the target definition into the model.

11.2 Class imbalance

The six categories do not occur equally often. Good and Severe days are much less common than Moderate and Satisfactory days. A classifier that predicts the most common class can achieve a nontrivial accuracy while failing to recognise rare categories. The project therefore uses balanced class weights in Logistic Regression, Decision Tree, Random Forest and HistGB where supported, tunes on macro-F1 and reports balanced accuracy.

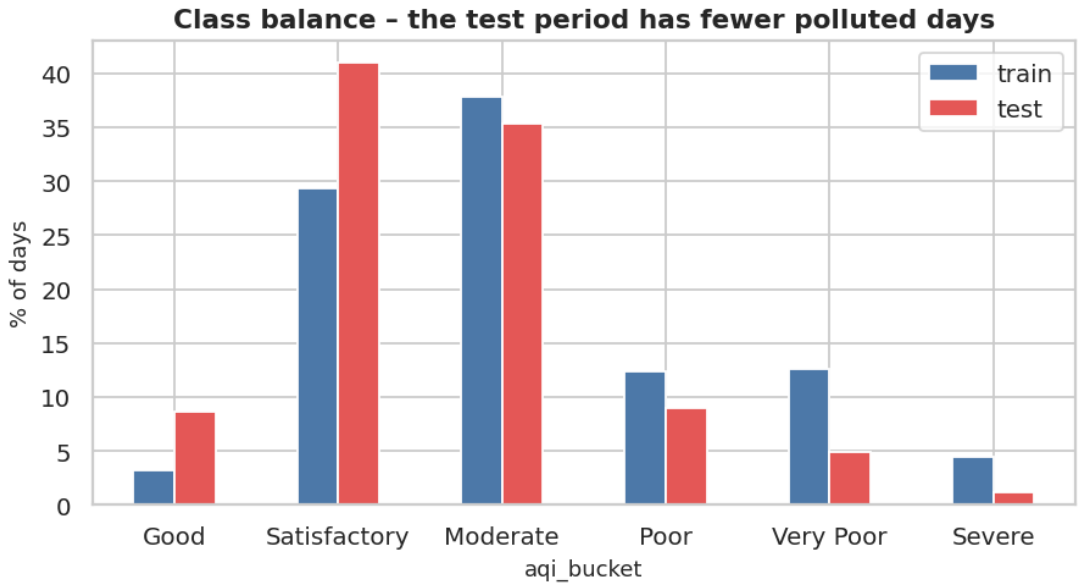


Figure 11.1: Class distribution in the same-day modelling sample. The imbalance motivates class-weighting and macro-averaged metrics.

11.3 Test results

Model	Accuracy	Balanced acc.	Macro-F1	Within ±1 class
Majority dummy	0.353	0.167	0.087	0.853
Logistic Regression	0.602	0.708	0.608	0.973
Decision Tree	0.726	0.759	0.703	0.991
HistGradientBoosting	0.761	**0.782**	0.733	0.996
MLP neural network	0.785	0.730	0.746	0.997
Random Forest	**0.785**	0.769	**0.756**	**0.997**

Random Forest is selected by training CV macro-F1 and attains the best test macro-F1 (0.756) and accuracy (0.785, tied with MLP). HistGB has the highest test balanced accuracy (0.782), suggesting slightly more even recall across classes. The right choice depends on which metric matters most.

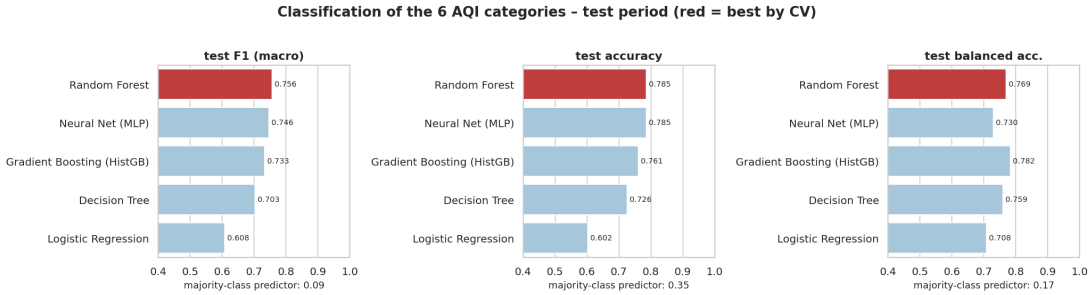


Figure 11.2: Classification model comparison. No single metric tells the whole story when the classes are imbalanced.

11.4 Confusion matrix: what kinds of errors occur?

A confusion matrix shows the actual label in rows and the predicted label in columns. Most predictions lie on or near the diagonal. In the Random Forest result, 78.5% of predictions are exactly correct; 21.2% are one category away; and only 0.3% are two categories away. No errors are more than two categories away in the reported distance summary.

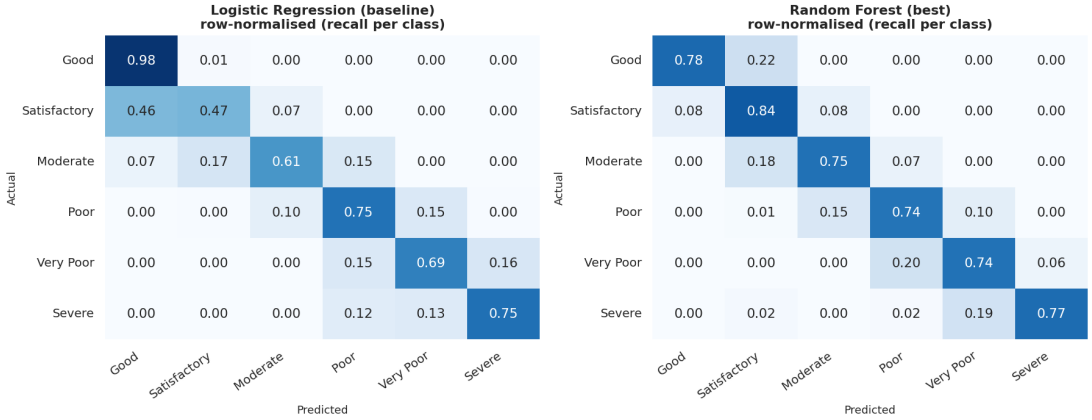


Figure 11.3: Confusion matrices for candidate classifiers. Adjacent-category confusion is more common than very large category jumps.

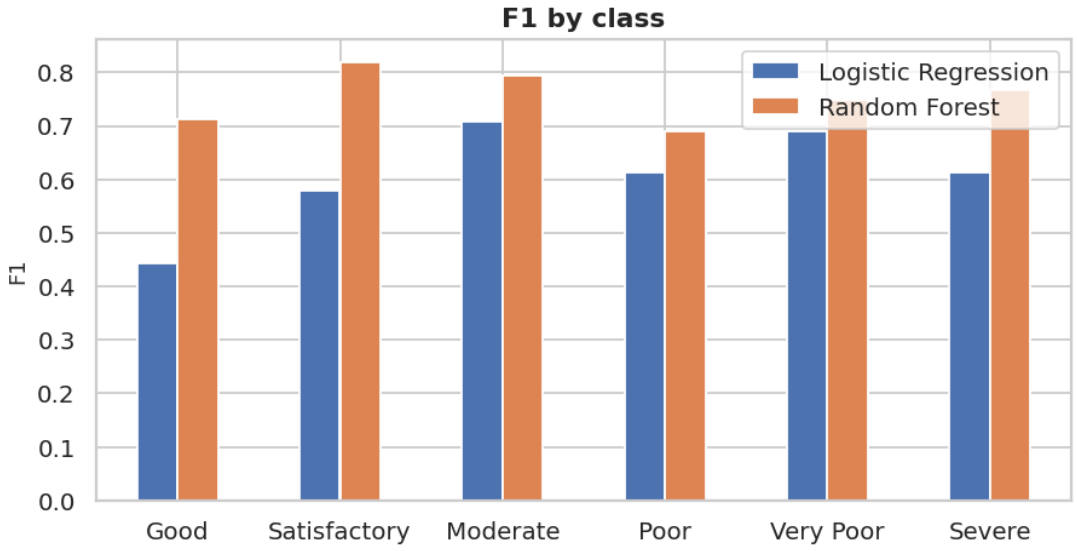


Figure 11.4: Per-class F1 scores expose performance differences hidden by overall accuracy.



Figure 11.5: Ordinal distance of classification errors. Since AQI bands are ordered, an adjacent-band error is less severe than a multi-band error.

11.5 Why direct classification can beat regression followed by bucketing

An alternative is to predict numeric AQI with a regressor and then convert that prediction into a category. The project compares this with directly learning the six category labels. Direct Random Forest classification performs better on macro-F1 and balanced accuracy than regression-then-bucket on the test set. This is reasonable because a regression model optimises numeric error, not the class boundaries or rare-class recall. A prediction that is numerically close but just across a threshold may have a category error.

11.6 Interpretation

A same-day classifier accuracy of around 78.5% means that about four out of five test labels are exact. The 99.7% within-one-class figure is encouraging, but it should not be mistaken for exact classification accuracy. For an operational health warning system, errors near a boundary need special attention, and calibrated probabilities or uncertainty intervals would be desirable.

Chapter 12

Next-day AQI forecasting: the more meaningful prediction task

12.1 Forecast target and information boundary

The next-day task predicts AQI_{t+1} using information available on date t and earlier. The available inputs include today's pollutant readings, today's AQI and its lags/rolling summaries, city, and calendar/event flags for tomorrow. Tomorrow's pollutant concentrations are not used because they would not yet be known at forecast time. Tomorrow's calendar is allowed because dates and festival schedules are known in advance.

Example. On 14 October, the model can use the pollutant readings and AQI observed on 14 October, plus the previous seven days of AQI and the fact that 15 October falls in the defined stubble-season window. It cannot use the PM2.5 measurement or AQI actually observed on 15 October.

Rows are excluded from training if their target date falls in the test period. The training feature dates therefore end at 30 December 2018 so that the target is no later than 31 December 2018. Test feature dates run from 1 January 2019 to 30 June 2020, with target dates extending to 1 July 2020. The next-day modelling split has 12,235 training rows and 10,779 test rows.

12.2 Persistence baseline

Persistence predicts that tomorrow will be the same as today:

$$\widehat{\text{AQI}}_{t+1} = \text{AQI}_t.$$

This is a strong baseline when daily pollution is temporally correlated. A model should not be called useful for forecasting merely because its R^2 is high; it should improve on persistence.

12.3 Next-day regression results

Model	MAE	RMSE	R^2	MAE skill vs persistence
Persistence	21.26	37.23	0.802	–
Linear Regression	18.91	29.88	0.873	11.1%
Decision Tree	19.30	30.22	0.870	9.2%
Random Forest	**17.12**	28.15	0.887	**19.5%**
HistGradientBoosting	17.29	**27.84**	**0.889**	18.7%
MLP neural network	21.72	32.35	0.851	-2.2%

MAE skill is defined as $1 - \text{MAE}_{model}/\text{MAE}_{persistence}$. Random Forest reduces MAE by approximately 19.5%. HistGB has a marginally higher R^2 and lower RMSE, while Random Forest has the lowest MAE and the best training CV MAE. Again, the models are close; the central result is that tree ensembles beat persistence, while the MLP does not.

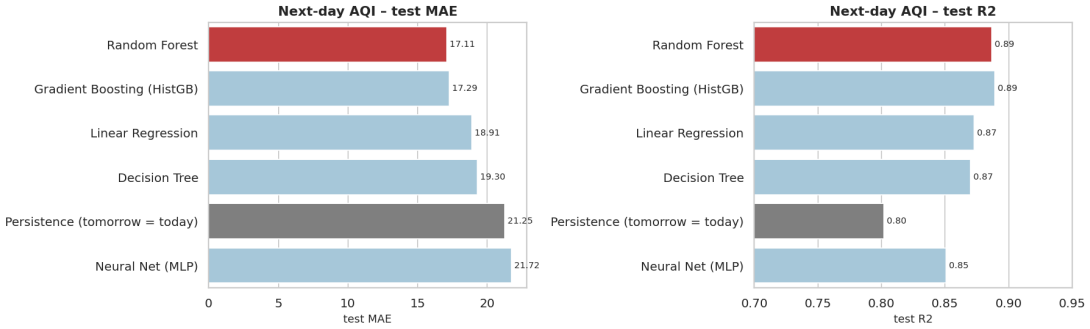


Figure 12.1: Next-day regression test results compared with persistence. The key measure is improvement over the persistence baseline.

12.4 Why pollutant readings matter more than AQI history alone

The ablation experiment progressively adds information to the forecasting model. AQI today alone, or AQI history/lag variables alone, do not beat persistence. Adding today's detailed pollutant readings substantially improves prediction. This is an important insight: today's AQI compresses several pollutant concentrations into one maximum sub-index. That compression discards information about pollutants that are not dominant today but may influence tomorrow's index if conditions change.

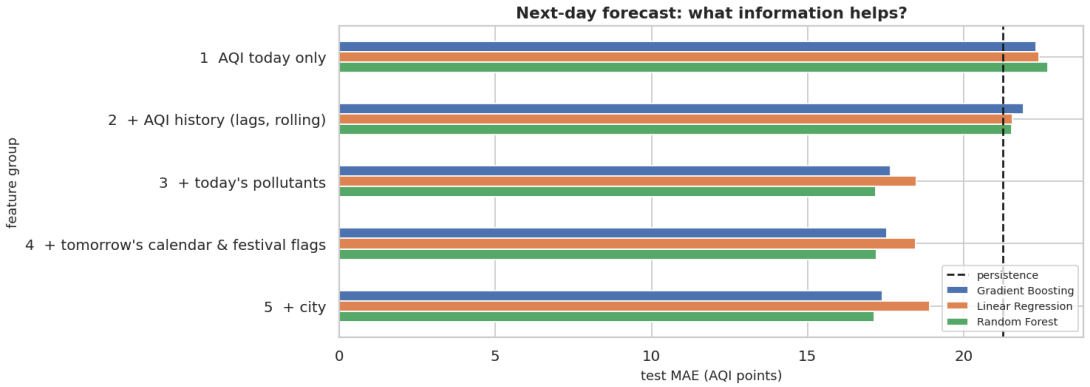


Figure 12.2: Ablation experiment: adding present-day pollutant readings produces the main improvement over persistence; calendar flags provide a smaller increment.

12.5 Feature importance for next-day prediction

The HistGB feature-importance analysis attributes the largest permutation effect to PM2.5 (MAE increase about 22.48 when shuffled), followed by today's AQI (+10.91), PM10 (+4.57) and the 7-day AQI rolling mean (+3.81). The model uses both current pollutant composition and recent AQI history.

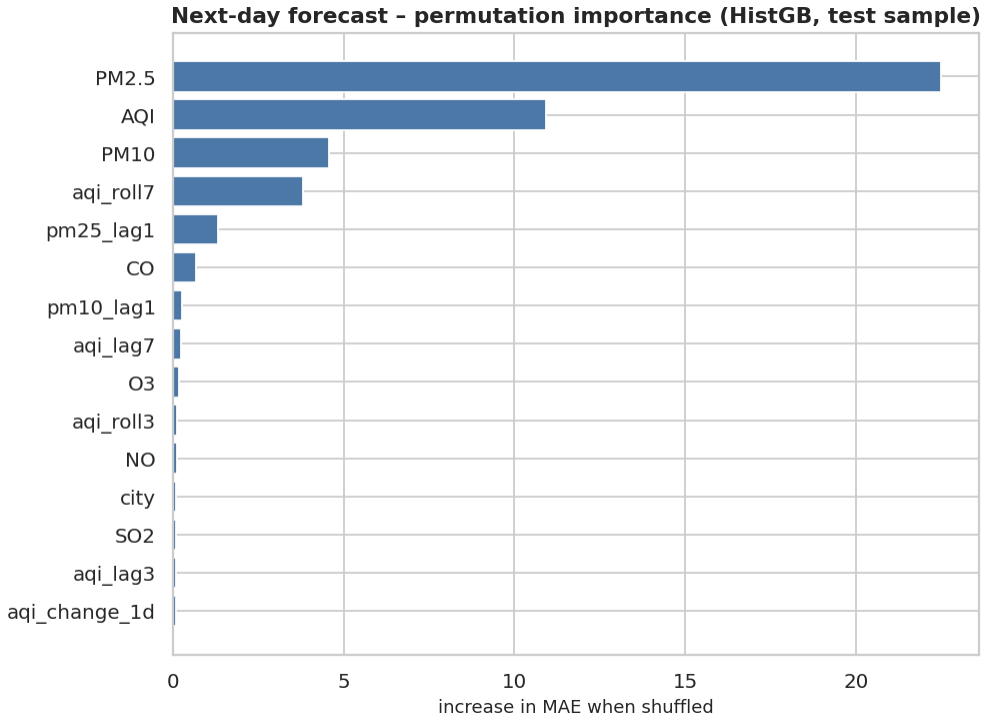


Figure 12.3: Permutation importance for next-day forecasting. Today's PM2.5 is the strongest individual input in this test.

12.6 Next-day classification

The next-day bucket classifier predicts tomorrow's AQI category. Random Forest achieves accuracy 0.774, balanced accuracy 0.758 and macro-F1 0.746, compared with persistence accuracy 0.737 and macro-F1 0.676. HistGB has slightly higher balanced accuracy (0.768) but lower macro-F1 (0.729). Logistic Regression and Decision Tree do not beat persistence in macro-F1 in the reported test.

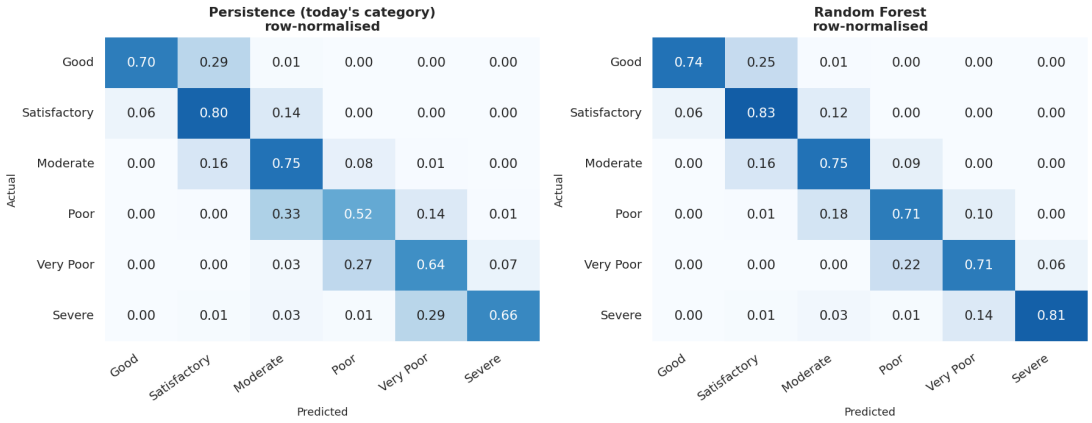


Figure 12.4: Next-day classification confusion matrix. The model frequently gets the adjacent category right even when exact classification fails.

12.7 City-level forecasting skill

Performance is not uniform by city. In the Random Forest test results, Delhi MAE improves from 35.51 for persistence to 27.71 for the model (about 22% skill); Chandigarh improves from 19.40 to 14.52 (about 25%); Mumbai from 13.35 to 11.08 (about 17%); and Bengaluru from 8.94 to 7.74 (about

13%). Some smaller-coverage cities do not show positive skill. Aggregate metrics can hide these differences.

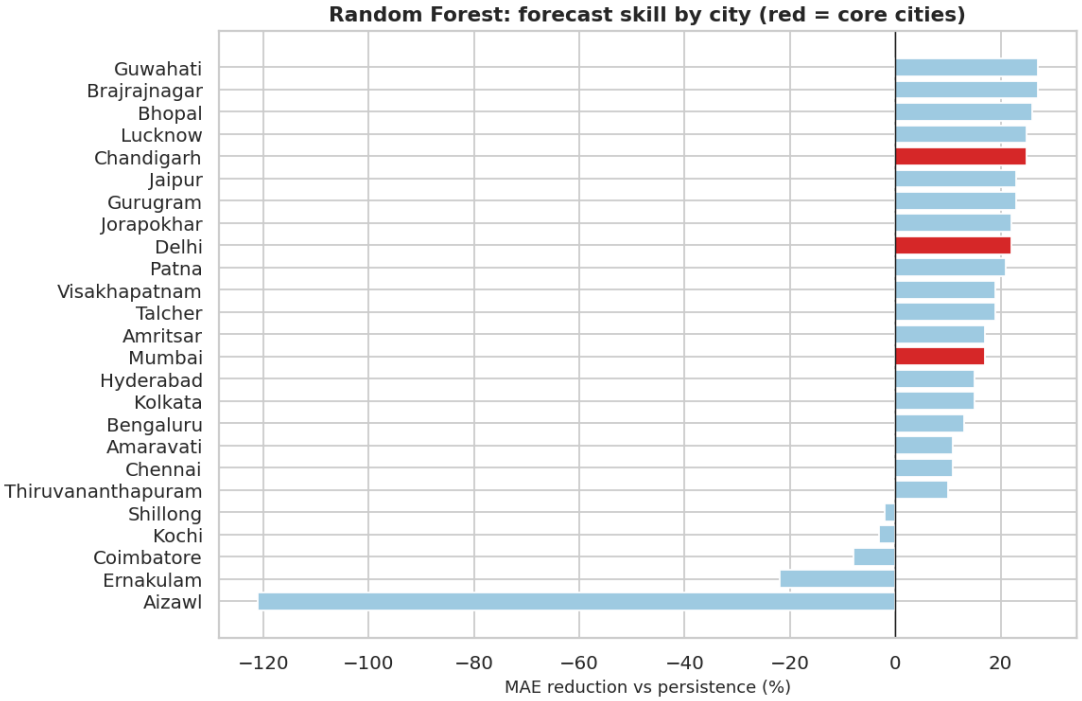


Figure 12.5: Next-day improvement over persistence varies by city; a pooled score should not be assumed to apply equally everywhere.

12.8 Diwali as a known-ahead event

Festival flags add little to average next-day performance across the full test set, but they can help around specific event windows. For the North India Diwali target-day sample (n=30), the selected Random Forest has MAE about 37.7 compared with persistence MAE about 53.5. A HistGB comparison without tomorrow event flags has MAE 43.3 versus 40.5 with the flags. This is a small event-specific sample, so it is indicative rather than conclusive.

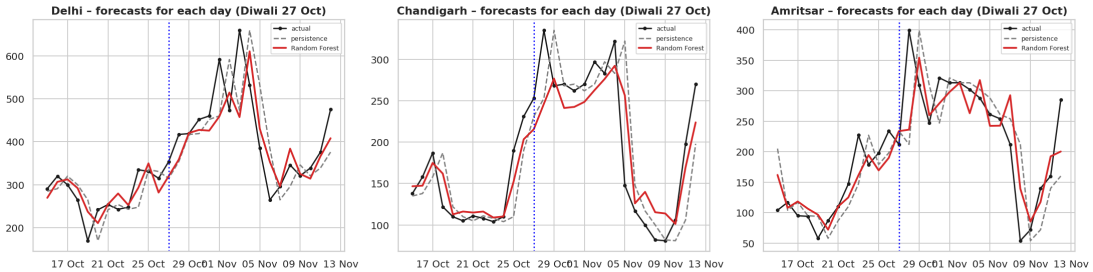


Figure 12.6: Diwali next-day case study. A known-ahead event flag can help on event days even if the average gain over all dates is small.

Chapter 13

City comparison and transfer analysis

13.1 Why use a common comparison window?

The full dataset has unequal city coverage. The direct city comparison therefore uses a shared interval from 2 September 2019 to 1 July 2020, chosen because Chandigarh's available time series begins in this period. This gives approximately 299–304 valid AQI days per city for the eight selected cities: Delhi, Chandigarh, Mumbai, Amritsar, Gurugram, Lucknow, Jaipur and Bengaluru.

13.2 City scorecard

City	Mean	Median	P90	Poor+ %	Main driver
Delhi	216.3	174.0	380.1	44.7	PM10
Chandigarh	96.5	82.0	171.8	7.7	PM10
Mumbai	109.6	88.5	192.0	8.2	PM10
Amritsar	112.5	91.0	202.2	10.2	PM10
Gurugram	186.0	153.0	338.5	31.9	PM2.5
Lucknow	185.1	157.5	343.7	38.8	PM2.5
Jaipur	114.0	108.0	161.4	3.0	PM10
Bengaluru	81.1	78.0	103.7	0.0	PM10

Mean AQI is highest in Delhi, followed by Gurugram and Lucknow. Chandigarh's mean is 96.5, close to Mumbai's 109.6 and well below Delhi's 216.3 during this common interval. The dominant pollutant in this window is PM10 for several cities, whereas PM2.5 is dominant for Gurugram and Lucknow. This period-specific driver differs from a full-history dominance summary; the dominant pollutant can change with city and time window.

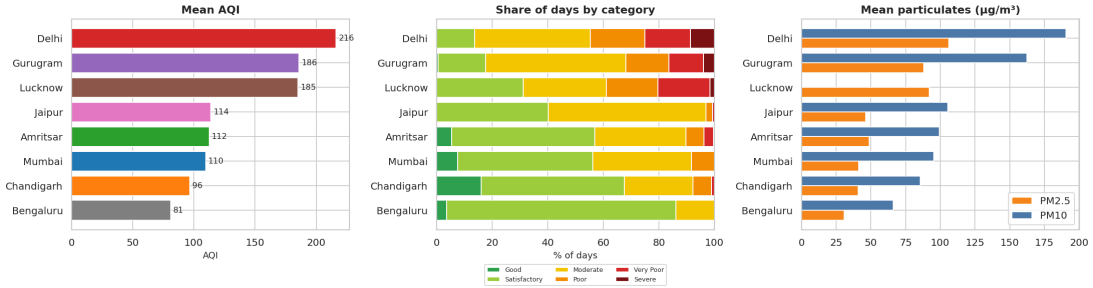


Figure 13.1: Like-for-like AQI level comparison for the eight cities over the common date window.

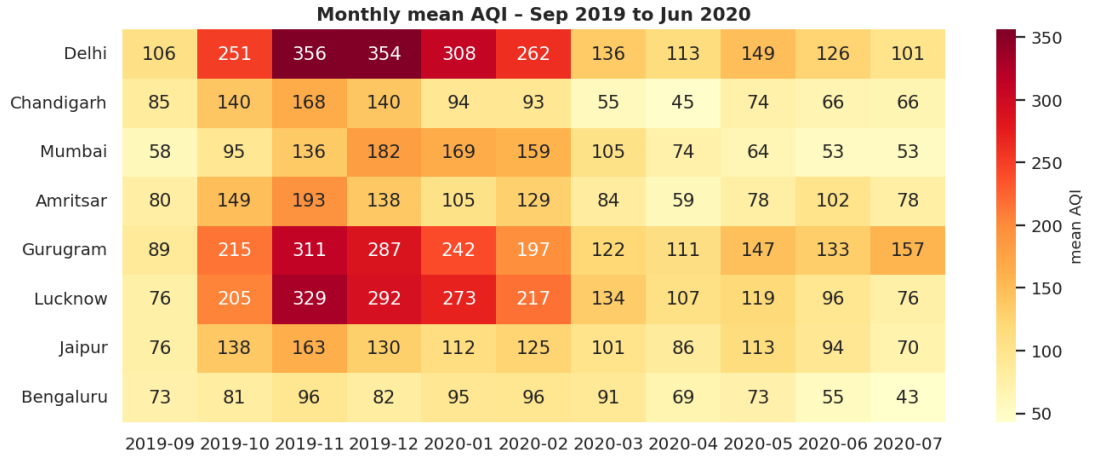


Figure 13.2: Monthly city profiles over the common period. Monthly differences help distinguish persistent city-level differences from seasonal patterns.

13.3 Do nearby cities move together?

Daily AQI correlations are high between Delhi and Gurugram (about 0.96), Delhi and Lucknow (0.86), Delhi and Chandigarh (0.79), and Chandigarh and Amritsar (0.73). Chandigarh has lower correlation with Mumbai (0.30) and Bengaluru (0.15). Most cross-city relationships peak at lag zero, suggesting broad synchronous co-movement during the observation period.

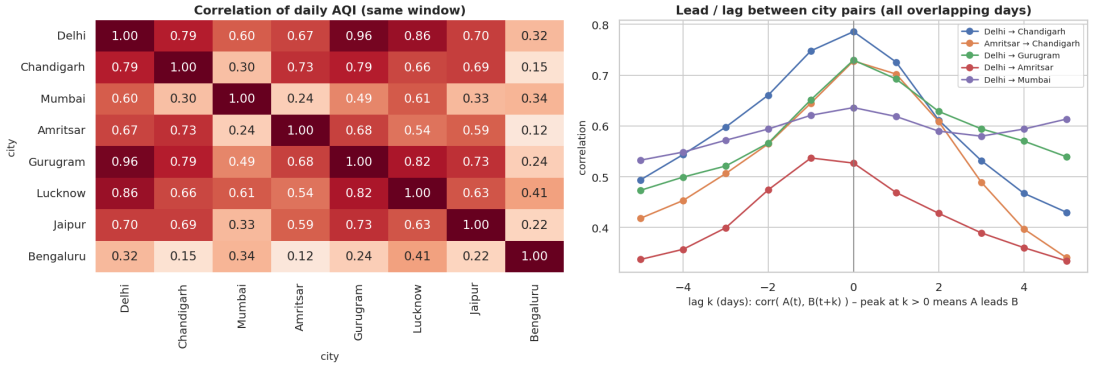


Figure 13.3: Cross-city daily AQI correlation. Correlation indicates shared movement, not proof that one city's pollution caused another's.

Correlation is not evidence of atmospheric transport by itself. Similar seasonal weather, regional emission sources and common policy/events can produce correlated time series. Wind direction and meteorological transport data would be needed to investigate causal movement of pollution between cities.

13.4 Pooled models versus city-specific models

A pooled model combines training observations from multiple cities, allowing it to learn common pollutant-AQI relationships. A local model uses only one city's own observations. In the 1 February–1 July 2020 comparison, Chandigarh's pooled model MAE is about 9.12 compared with 14.49 for a local model; a pooled model trained without Chandigarh achieves about 8.95 on Chandigarh (zero-shot). This is promising evidence of cross-city transfer, but the local Chandigarh model has only about 148 training rows in that particular comparison, making the comparison data-limited.

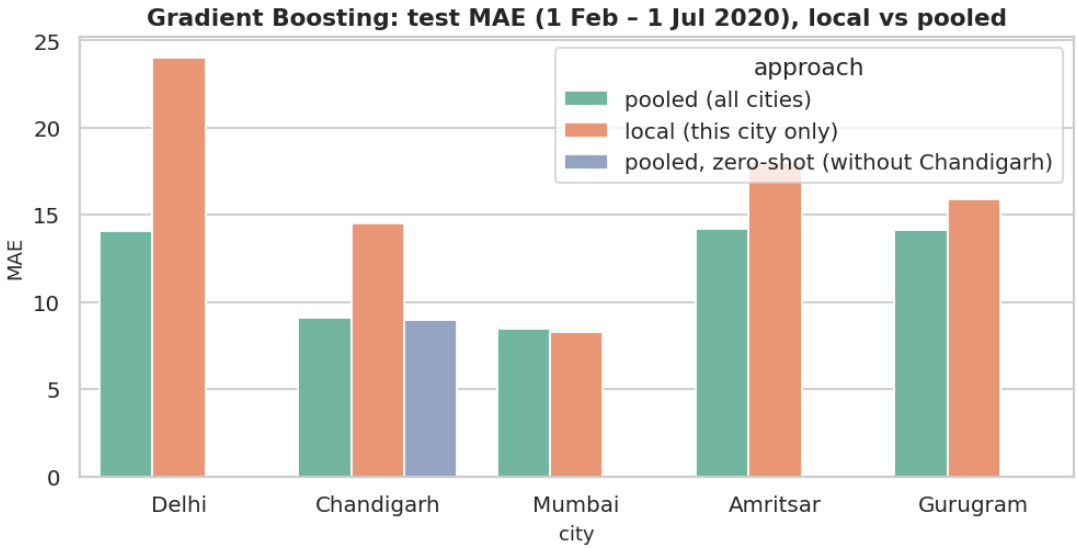


Figure 13.4: Pooled-versus-local city model comparison. A pooled model can use a much larger training set than a city-specific model.

13.5 City-level Diwali and lockdown differences

Diwali 2019 increases AQI sharply in Amritsar (+139.8%), Chandigarh (+124.3%), Jaipur (+71.7%), Delhi (+59.1%) and Gurugram (+46%) in the selected comparison window; Mumbai changes by approximately -2.6%. Lockdown comparisons show declines across several cities, including Lucknow (-45.3%), Delhi (-44.4%) and Gurugram (-41.9%). The Chandigarh same-date prior-year comparison is unavailable because the city has no corresponding 2019 earlier-year record in its short coverage window. A pre-lockdown comparison for Chandigarh is possible but confounded by seasonal drift.

Chapter 14

Results synthesis: what should we conclude?

14.1 The central findings

1. **The database and cleaning system works end-to-end.** The repository builds a relational SQLite database with city and fact tables, a cleaning audit trail, analytical views and stored model results. Integrity checks show no foreign-key violations and no duplicate city-date rows.
2. **The data are uneven and imperfect.** Missingness differs by pollutant and year; city coverage is unbalanced; placeholder zero values require domain judgment; Ahmedabad contains strong anomalies.
3. **Season and geography matter.** Delhi and other northern cities have much higher observed AQI than many southern cities, and northern AQI peaks in the post-monsoon/winter period.
4. **Particulate matter drives much of AQI variation.** PM2.5 and PM10 show the strongest associations with AQI and are the most important same-day predictors.
5. **Four features can retain almost all same-day regression performance.** PM2.5, PM10, CO and O3 are a compact set selected through time-series CV.
6. **Same-day estimation is not the same as forecasting.** The AQI calculation itself depends on the pollutant readings, so same-day models partly learn an index formula.
7. **Next-day forecasting is a stricter test.** Tree ensembles reduce MAE by roughly 19% relative to persistence by using today's pollutant composition and recent history.

14.2 Performance numbers to remember

Task	Baseline	Strong model	What it means
Same-day regression	Linear MAE 22.37	RF MAE 15.96; MLP R^2 0.913	Good estimation of reported AQI from same-day pollutant measurements.
Same-day classification	Logistic macro-F1 0.608	RF macro-F1 0.756; acc. 0.785	Most test categories are correct or one adjacent category away.
Next-day regression	Persistence MAE 21.26	RF MAE 17.12; skill 19.5%	Today's pollutant readings add forecasting value beyond today's AQI.
Next-day classification	Persistence macro-F1 0.676	RF macro-F1 0.746	Forecast category improves, though errors remain.

14.3 What the headline metrics do not say

MAE is an average: it does not guarantee a maximum error bound. R^2 depends on target variance, so a lower R^2 in a low-variance period can coexist with lower MAE. Accuracy can hide class imbalance, so macro-F1 and balanced accuracy are needed. A model tested on the same historical dataset is not automatically generalisable to future years, different sensor networks or different countries. Finally, association in EDA does not prove environmental causation.

Chapter 15

Limitations and concrete next steps

15.1 Limitations

1. **No meteorology.** Wind, rainfall, temperature, humidity and boundary-layer height can strongly influence pollutant concentration. These are the most important missing predictor families for a better forecast.
2. **Uneven city/time coverage.** Chandigarh has around ten months, Mumbai's AQI is largely absent, and some cities contribute much more history than others.
3. **Same-day target circularity.** Today's AQI is constructed from today's pollutant measurements; high same-day performance should not be sold as an advance warning forecast.
4. **Daily city aggregates.** The data do not support hourly forecasting or station-level spatial detail.
5. **Outlier and suspect-city decisions.** The Ahmedabad exclusion is reasoned but still a judgement call; alternate treatment should be reported in sensitivity analysis.
6. **Hard-coded calendar events.** Festival dates and event windows need validation before re-use, and event-study sample sizes can be modest.
7. **Lockdown confounding.** Before/after comparisons are observational and combine emissions changes, weather and seasonal effects.
8. **Limited temporal validation design.** A future version should use grouped date-based folds for all cities, nested feature selection and repeated backtesting across multiple cutoffs.
9. **No uncertainty intervals.** A single prediction does not communicate how uncertain a forecast is, especially for extreme AQI.

15.2 Recommended improvements in priority order

1. Add daily meteorological variables from a documented source, align by city and date, and assess missingness/coverage carefully.
2. Use a grouped time-series split where all cities from a date belong to the same fold; perform rolling-origin backtests over several forecast cutoffs.
3. Add prediction intervals, quantile regression or conformal intervals, especially for health-relevant high-AQI predictions.
4. Test station-level data or a spatial model if the research question requires within-city variation.
5. Compare one-, two- and three-day forecast horizons using persistence and seasonal baselines.

6. Improve source lineage: preserve the unmodified CSV and raw database import, then create separate staging and curated tables.
7. Investigate Ahmedabad's anomalies with the source provider before deciding whether it should be excluded or corrected.

Chapter 16

Reproducibility and repository guide

16.1 Notebook sequence

Notebook / module	Purpose
`01_database_pipeline.ipynb`	Creates and populates SQLite; runs cleaning SQL; demonstrates queries, updates, deletes, rollback and integrity checks.
`02_cleaning_and_features.ipynb`	Studies missingness, placeholder zeros, gap lengths, imputation and feature-table construction; validates the CPCB reconstruction.
`03_EDA.ipynb`	Produces distributions, city/region comparisons, seasonal profiles, correlations, pollutant drivers and event analyses.
`04_feature_reduction.ipynb`	Evaluates correlation filtering, VIF, importance, feature-count curves and PCA.
`05_regression_same_day.ipynb`	Fits/evaluates same-day numeric AQI regression and diagnostics.
`06_classification.ipynb`	Fits/evaluates six-class AQI category classification and ordinal errors.
`07_city_comparison_and_nextday.ipynb`	Compares selected cities, pooled/local models, next-day regression/classification, ablations and event-specific forecasting.

16.2 Reusable source modules

Module	Responsibility
`config.py`	Paths, pollutant lists, AQI bands, city metadata, festival dates, seasons and split dates.
`db.py` / `pipeline.py`	Database connections, SQL execution, pipeline orchestration and audit logging.
`features.py`	Calendar/event features, gap densification, limited forward filling, lag/rolling features and valid-row flags.
`cpcb.py`	Pollutant sub-index interpolation, reconstructed AQI and dominant-pollutant identification.
`modeling.py`	Preprocessing pipelines, model catalogue, chronological split and RandomizedSearchCV with TimeSeriesSplit.

`evaluate.py` / `viz.py`	Metric calculation and consistent diagnostic plotting.

16.3 Re-run instructions

From the repository root, create a virtual environment, install `requirements.txt`, then open Jupyter and execute notebooks 01–07 in order. Alternatively run `python run_pipeline.py` to rebuild the database and processed feature table without executing all notebooks. The shipped notebooks contain outputs, and the project notes that minor numerical differences may occur under different dependency versions. The optional XGBoost model is included only when the package is installed; the scikit-learn HistGradientBoosting model is available as the default gradient-boosted alternative.

```
python -m venv .venv
source .venv/bin/activate # Windows: .venv\\Scripts\\activate
pip install -r requirements.txt
jupyter lab # run notebooks 01 through 07

# Or rebuild the data pipeline only:
python run_pipeline.py

# Run the unit tests:
python -m unittest discover -s tests -v
```

The supplied 13 unit tests pass in the inspected environment. They check core data transformations and logic; passing unit tests does not prove that the scientific assumptions are correct, but it does provide regression protection for key code behaviour.

Chapter 17

Glossary

Term	Meaning in this project
AQI	Air Quality Index, a number derived from pollutant-specific sub-indices.
Sub-index	Index value calculated for one pollutant using concentration breakpoints.
Target	Variable the model is asked to predict.
Feature	Input variable used by the model.
Regression	Prediction of a continuous numeric value.
Classification	Prediction of a discrete class label.
EDA	Exploratory Data Analysis; understanding distributions, relationships and data quality.
ETL	Extract, Transform, Load: moving raw data into a structured destination after transformations.
Missingness	An observation not recorded or unavailable.
Imputation	Filling missing feature values using a defined rule or model.
Forward fill	Carrying the last available value forward in time.
Lag	A previous observation aligned with the current date.
Rolling mean	Average of a fixed trailing time window.
Data leakage	Use of information that would not be available at the time of prediction or evaluation.
Baseline	A simple reference model that a more complex model should beat.
Persistence	Forecast tomorrow by copying today's observed value.
MAE	Mean Absolute Error; typical absolute prediction error in target units.
RMSE	Root Mean Squared Error; penalises large errors more than MAE.
R^2	Proportion of target variance explained relative to a mean predictor.
Macro-F1	Unweighted average of per-class F1 scores.
Balanced accuracy	Average recall across classes.
Confusion matrix	Table of actual versus predicted class counts.
VIF	Variance Inflation Factor; diagnostic for linear multicollinearity.
PCA	Principal Component Analysis; maps inputs into orthogonal variance-ranked components.
Permutation importance	Performance change after randomly shuffling one feature.
One-hot encoding	Converts a categorical variable into binary indicator columns.
TimeSeriesSplit	Cross-validation that validates on later time blocks than training blocks.

Chapter 18

References and technical resources

1. Central Pollution Control Board (CPCB), Government of India, *National Air Quality Index* and associated technical report. The project implements the pollutant sub-index interpolation and maximum-sub-index aggregation logic described in the national AQI methodology. https://app.cpcbccr.com/ccr_docs/FINAL-REPORT_AQI_.pdf
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4. F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. https://jmlr.org/papers/v12/pedregosa11a.html
5. Scikit-learn documentation: pipelines, model selection, TimeSeriesSplit, metrics, tree ensembles and preprocessing. https://scikit-learn.org/stable/